Diabetes Analysis with Bayesian Network

Fundamentals Of AI - Module 3
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Spring 2026

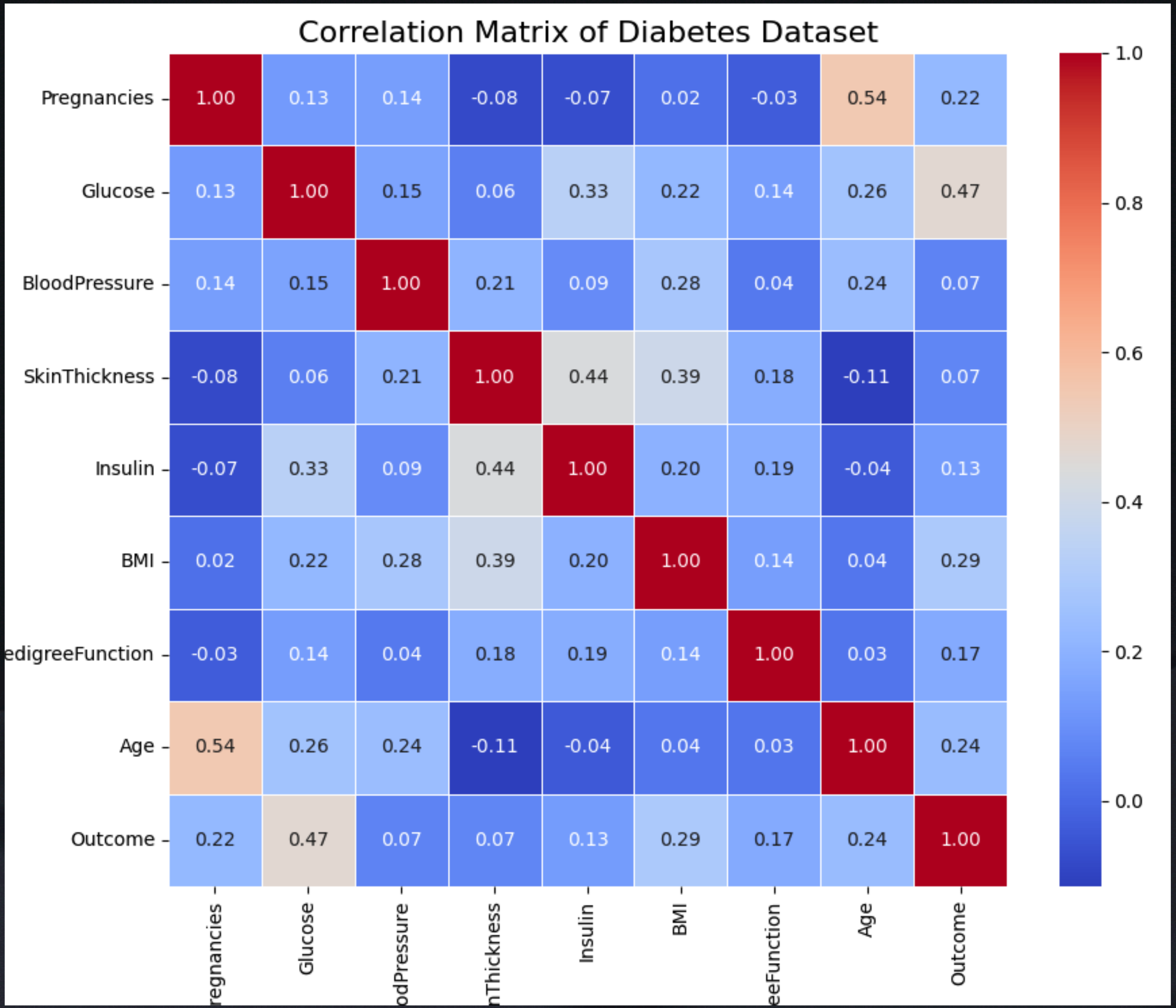
Dataset Overview

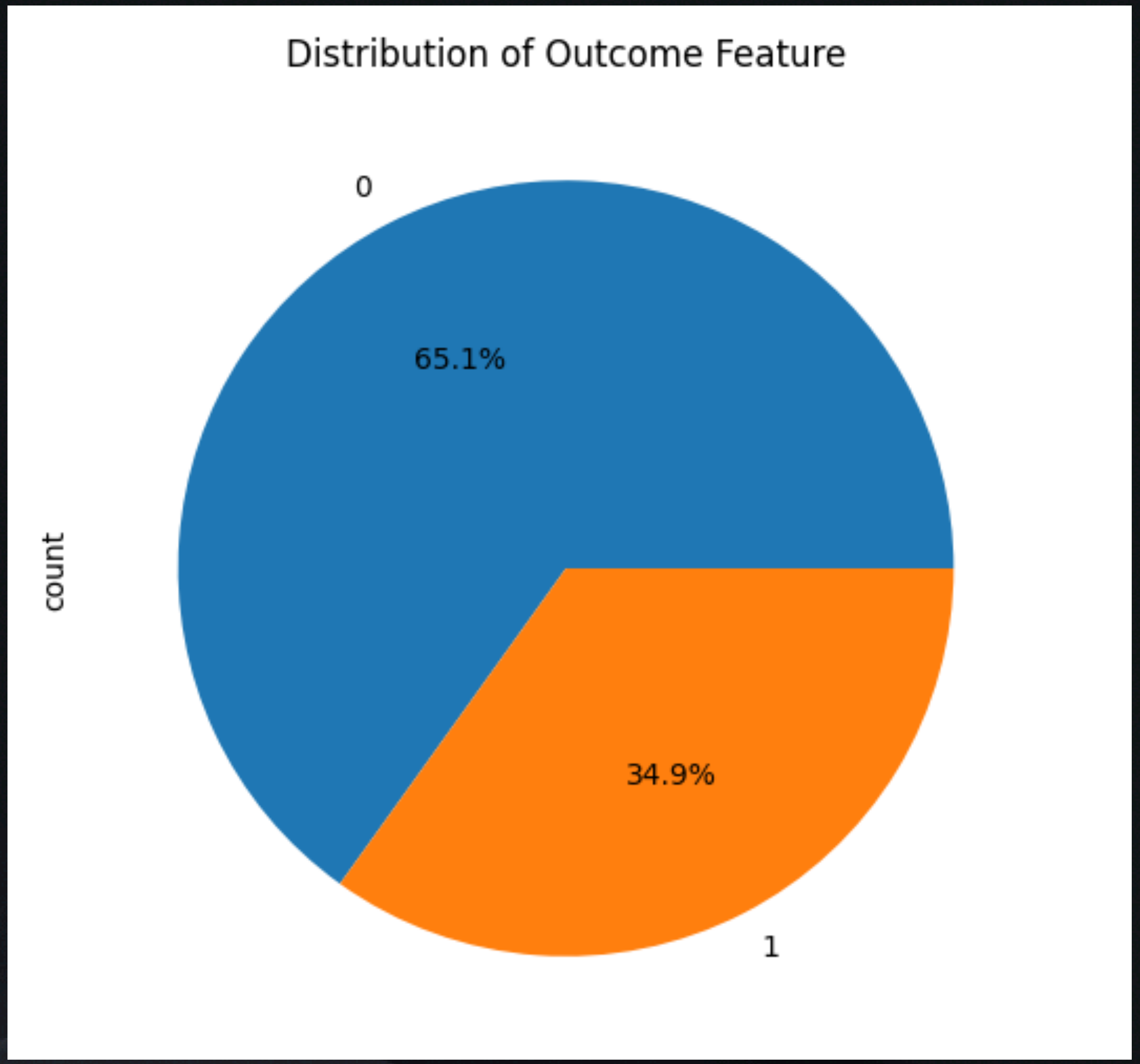
Attribute	*Description*
Pregnancies	Number of times pregnant
Glucose	Plasma glucose concentration 2 hours after an oral glucose tolerance test
BloodPressure	Diastolic blood pressure (mm Hg)
SkinThickness	Triceps skin fold thickness (mm)
Insulin	2-hour serum insulin (μU/ml)
BMI	Body mass index (weight in kg / height in m^2)
DiabetesPedigreeFunction	Diabetes pedigree function
Age	Age (years)
Outcome	Class variable (0 or 1) indicating if the patient has diabetes (1) or not (0)

Exploratory Data Analysis

- statistical summary of the dataset
- plotting distributions of features
- checking for missing values
- checking for zero values in features where it is not possible
- scatter plot of each feature vs Outcome
- pair plot of features
- heatmap of correlation matrix

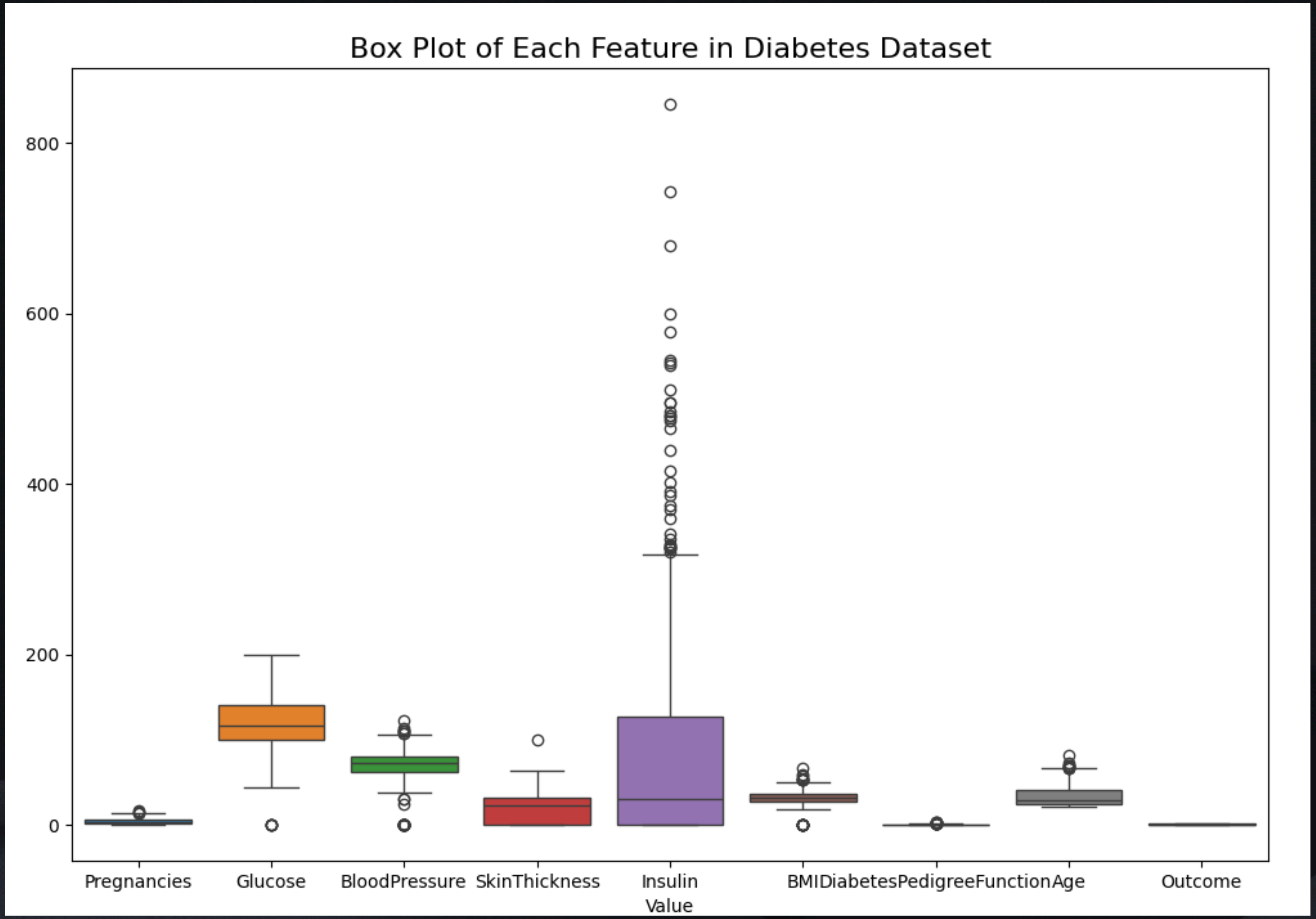
Data Sourcing

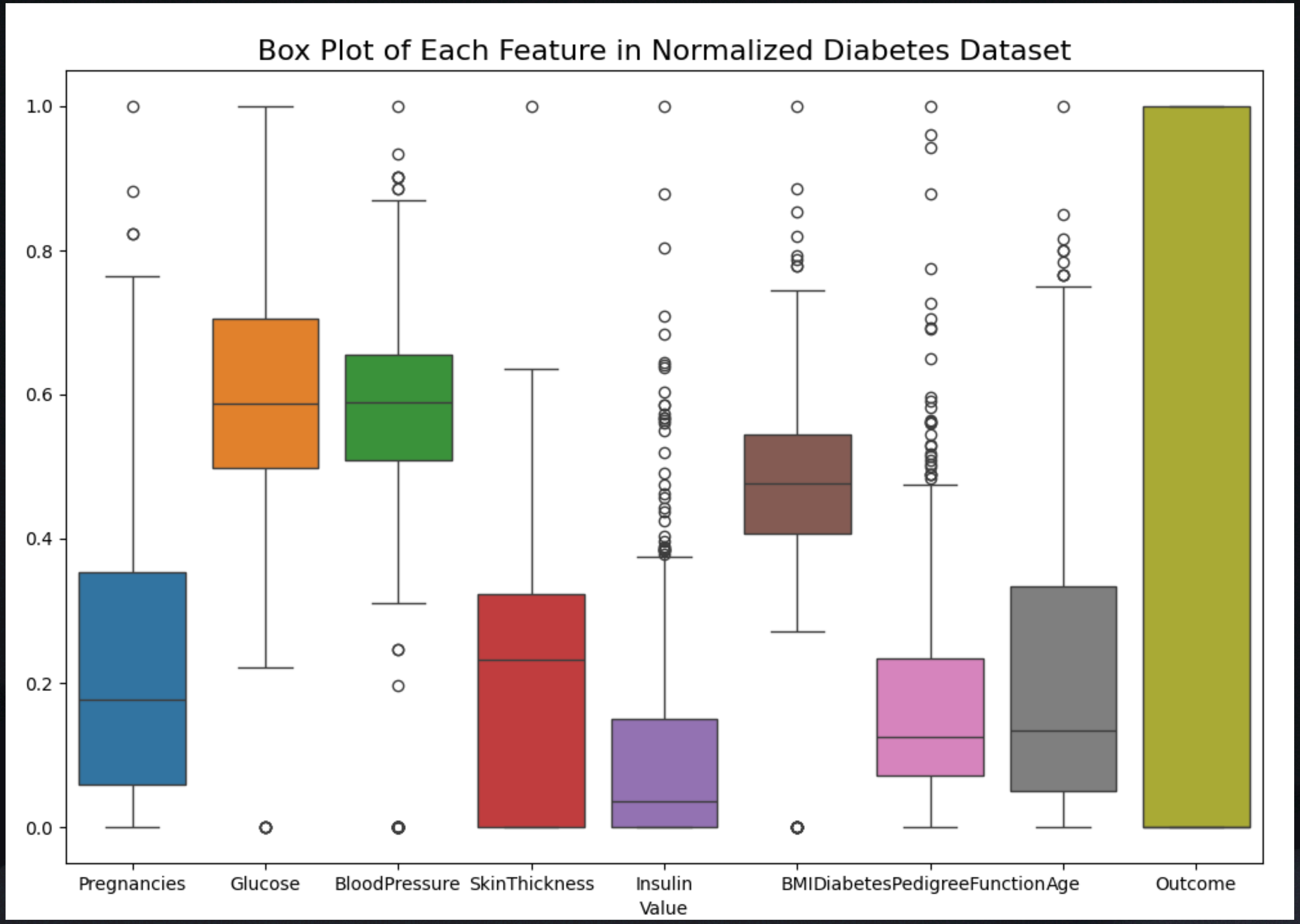




- 768 total instances
- 500 non-diabetic (0), 268 diabetic (1)
- Mix of continuous and categorical features

Normalize (MinMax Scaler)





Preprocessing

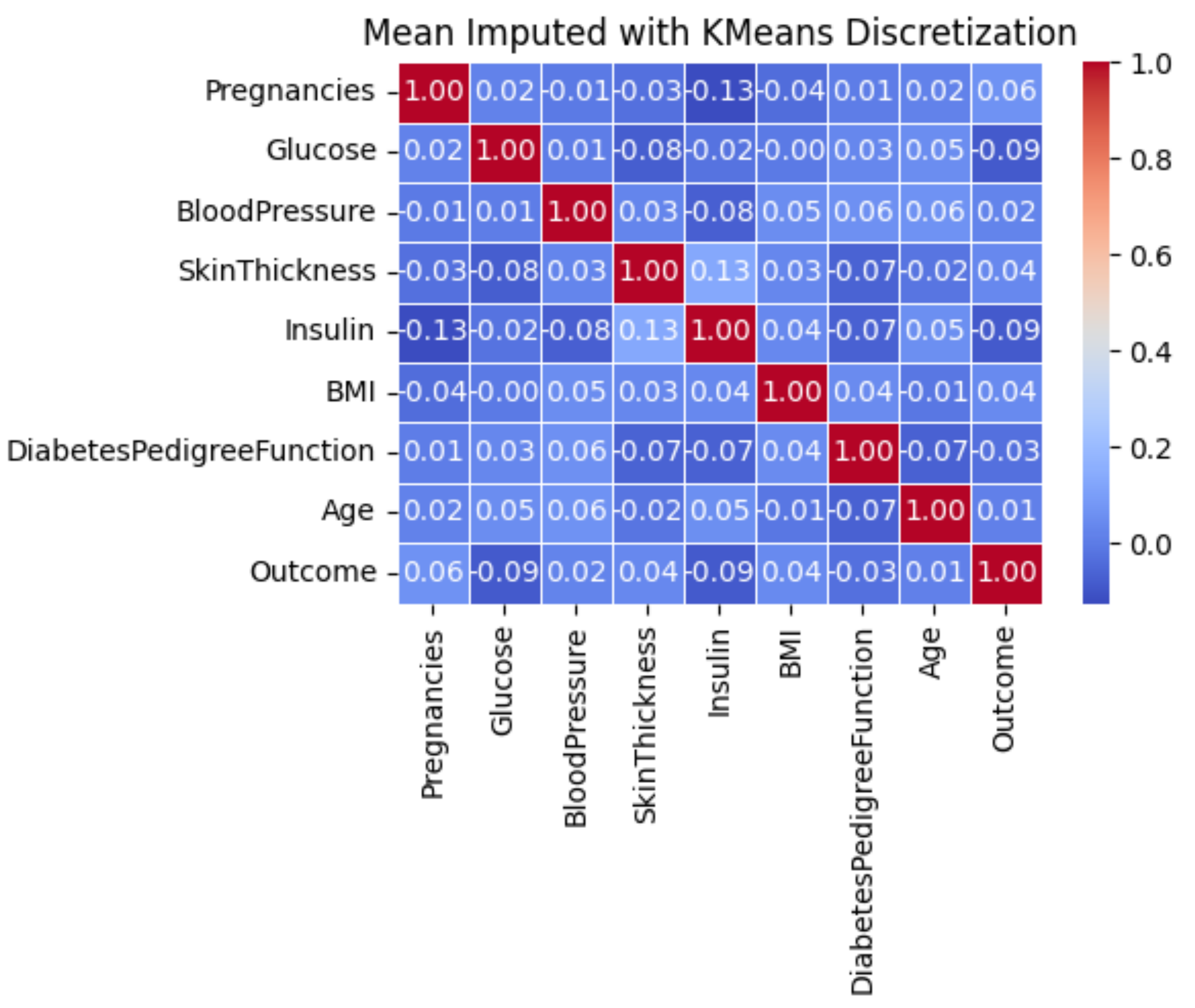
- *Handel Zero Values*
  - Remove rows with missing values
  - Impute missing values with mean/median/mode
  - Use advanced imputation techniques like KNN imputation or regression imputation

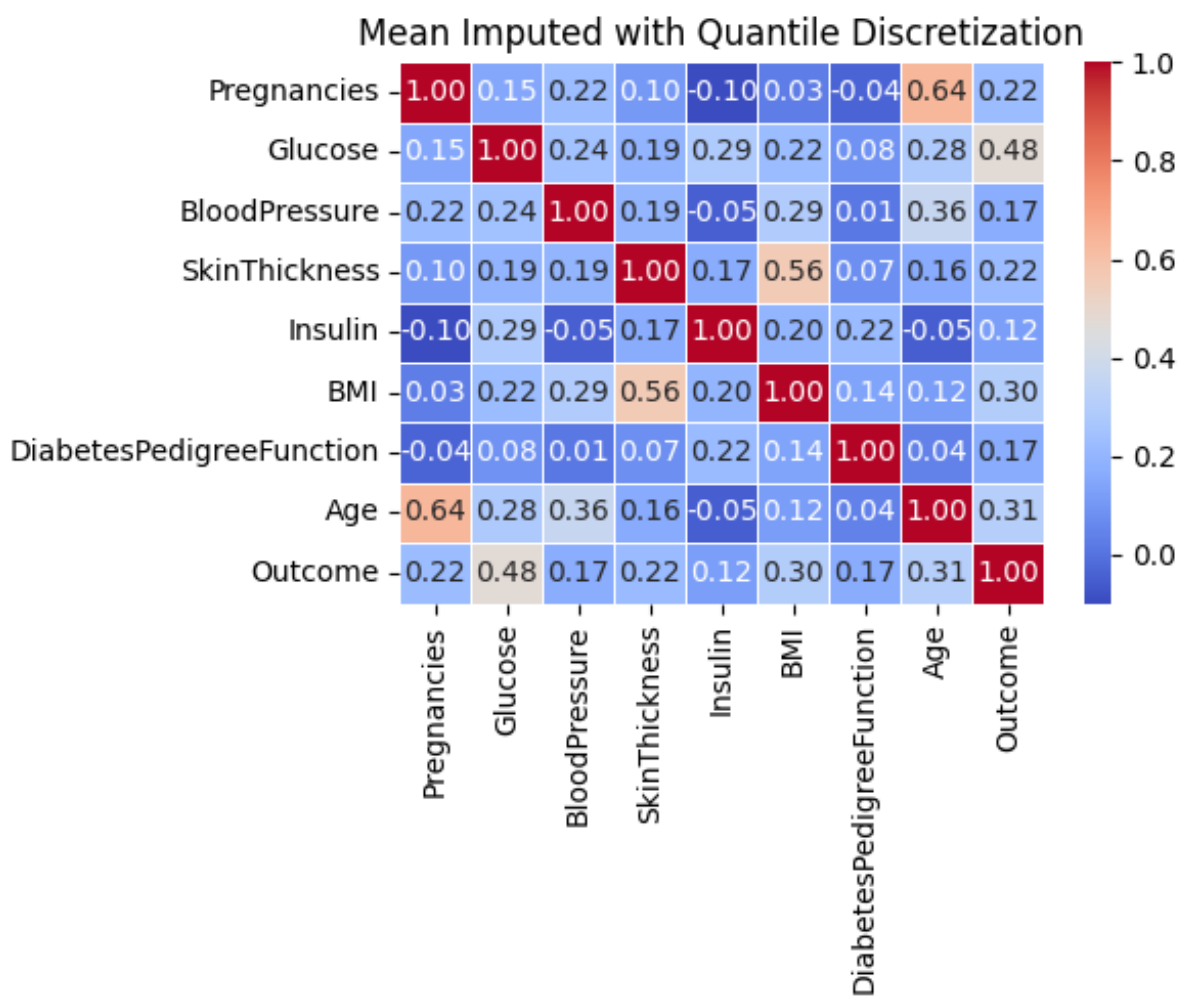
{'Glucose': 5, 'BloodPressure': 35, 'SkinThickness': 227, 'BMI': 11, 'Age': 0}

Preprocessing

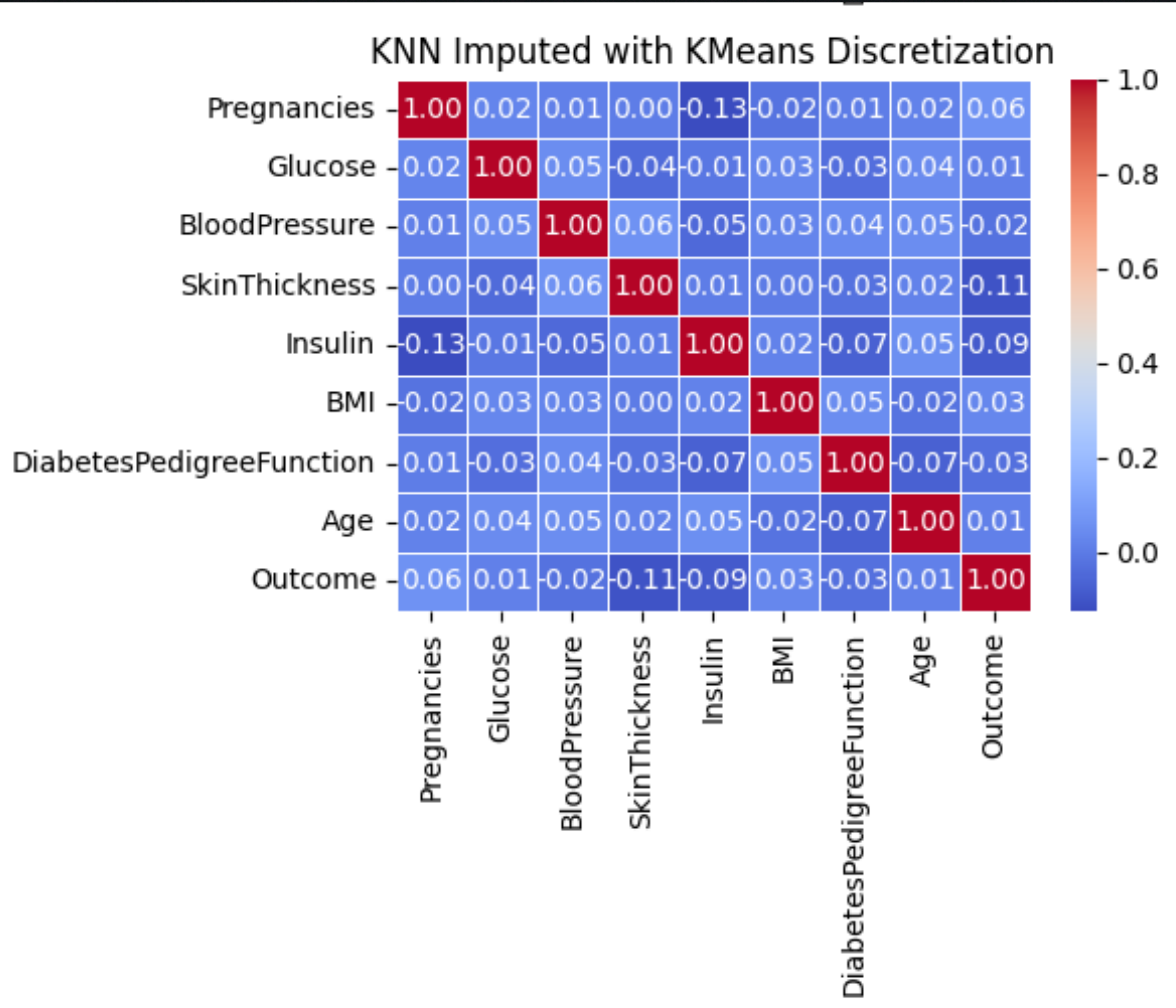
- *Handel Discretization*
  - Equal Width Binning
  - Quantile Binning
  - K-Means Binning
  - Round Values to nearest integer

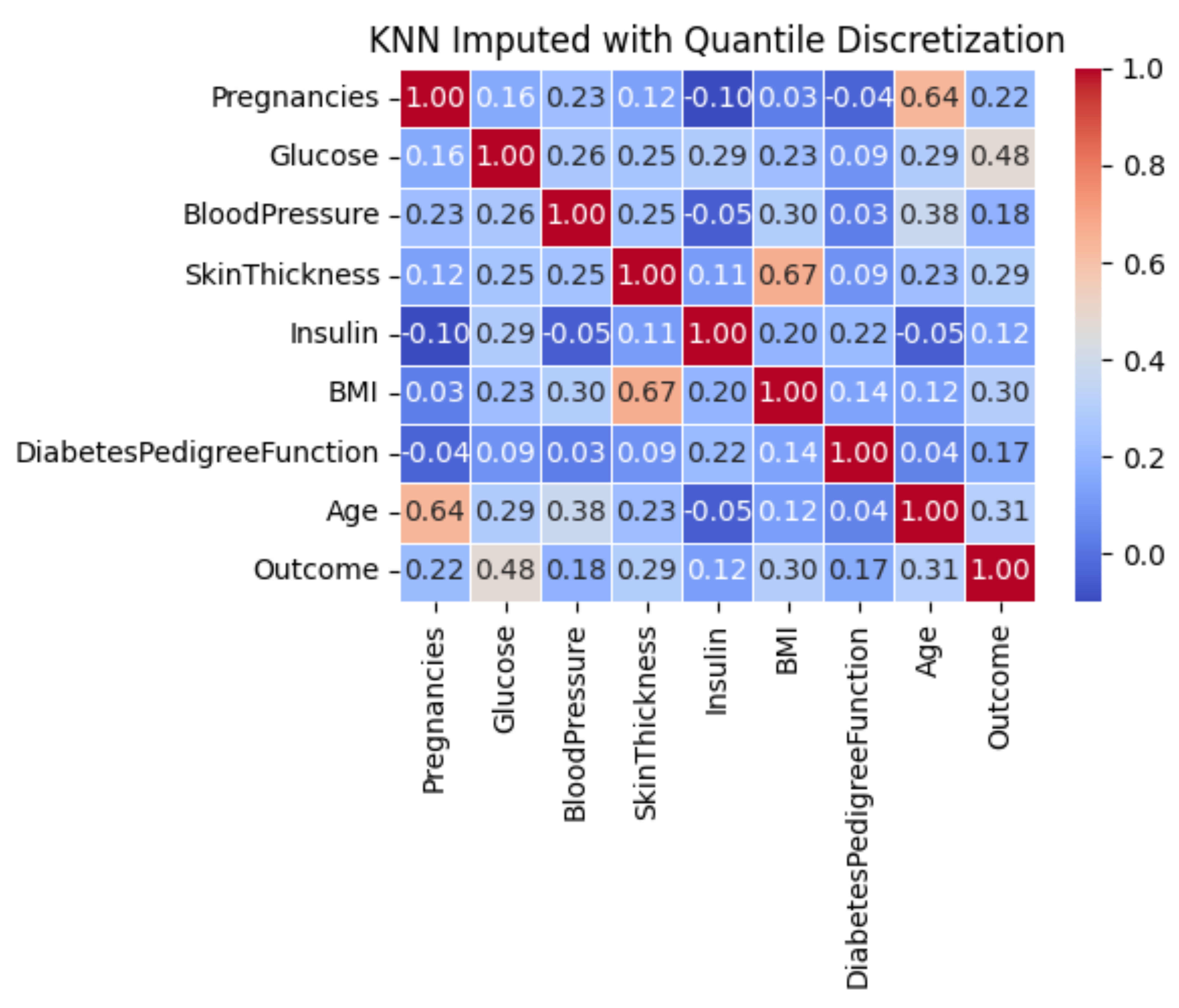
Correlation Matrix (Mean)



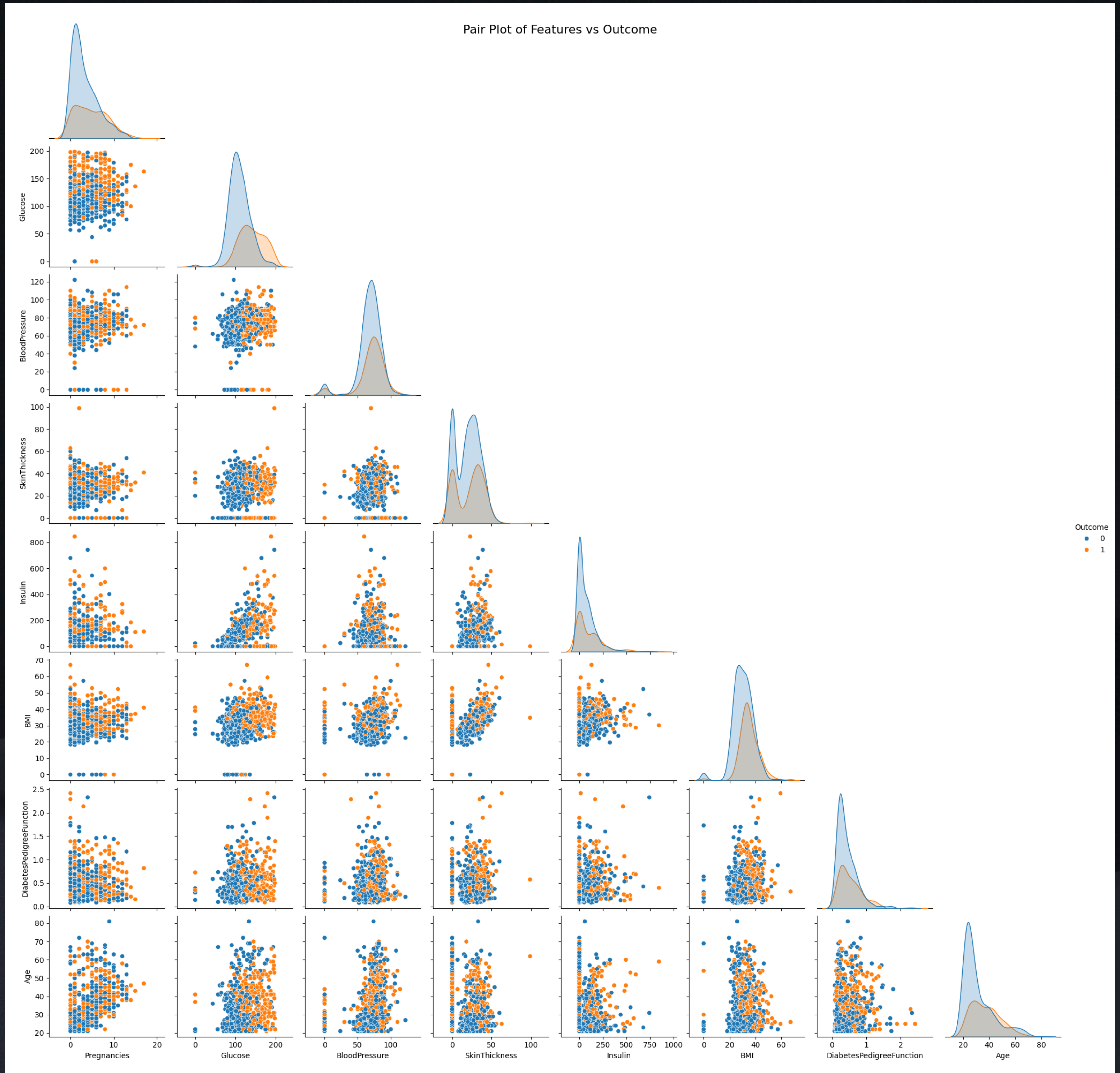


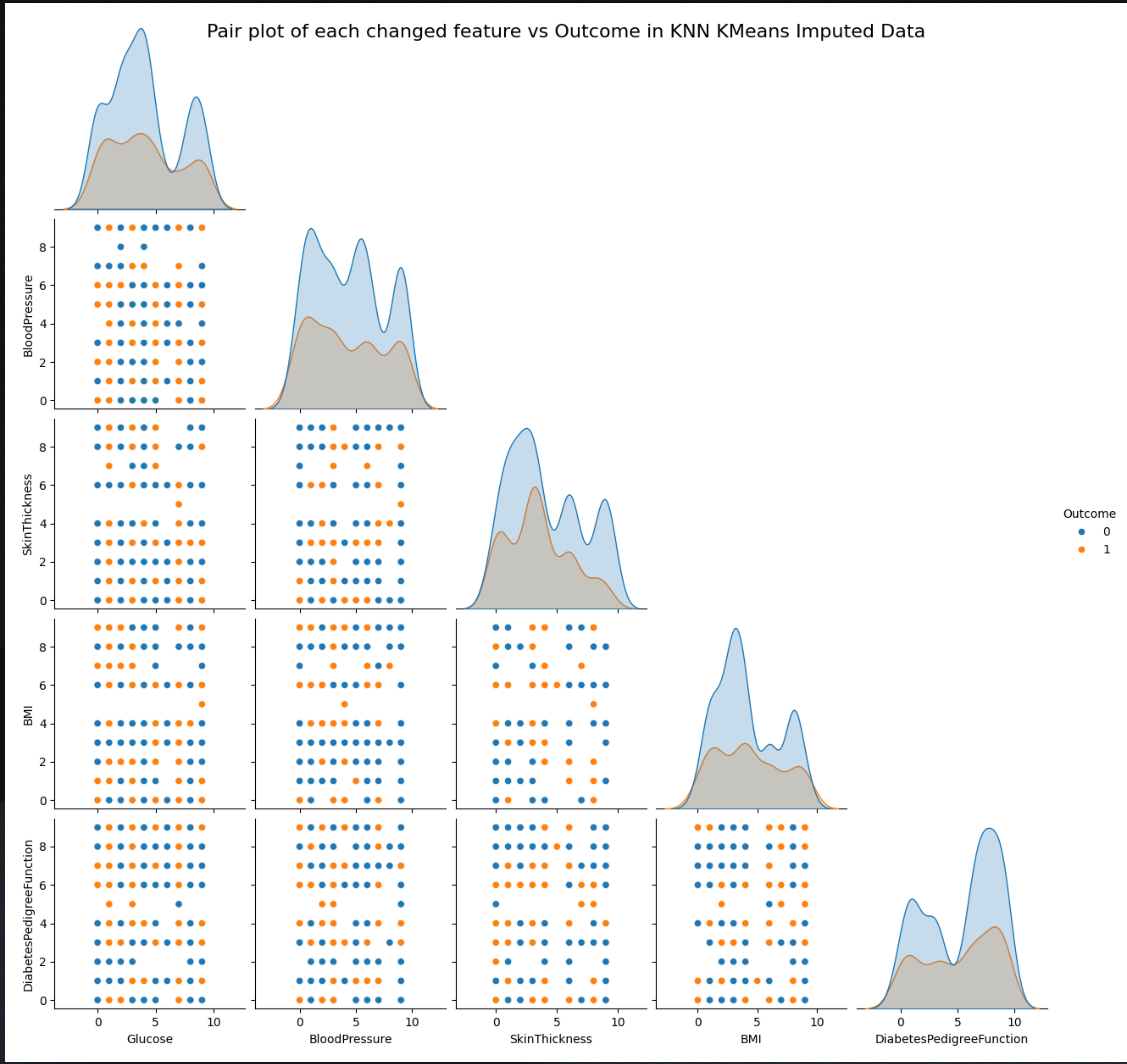
Correlation Matrix (KNN)





Pair Plot

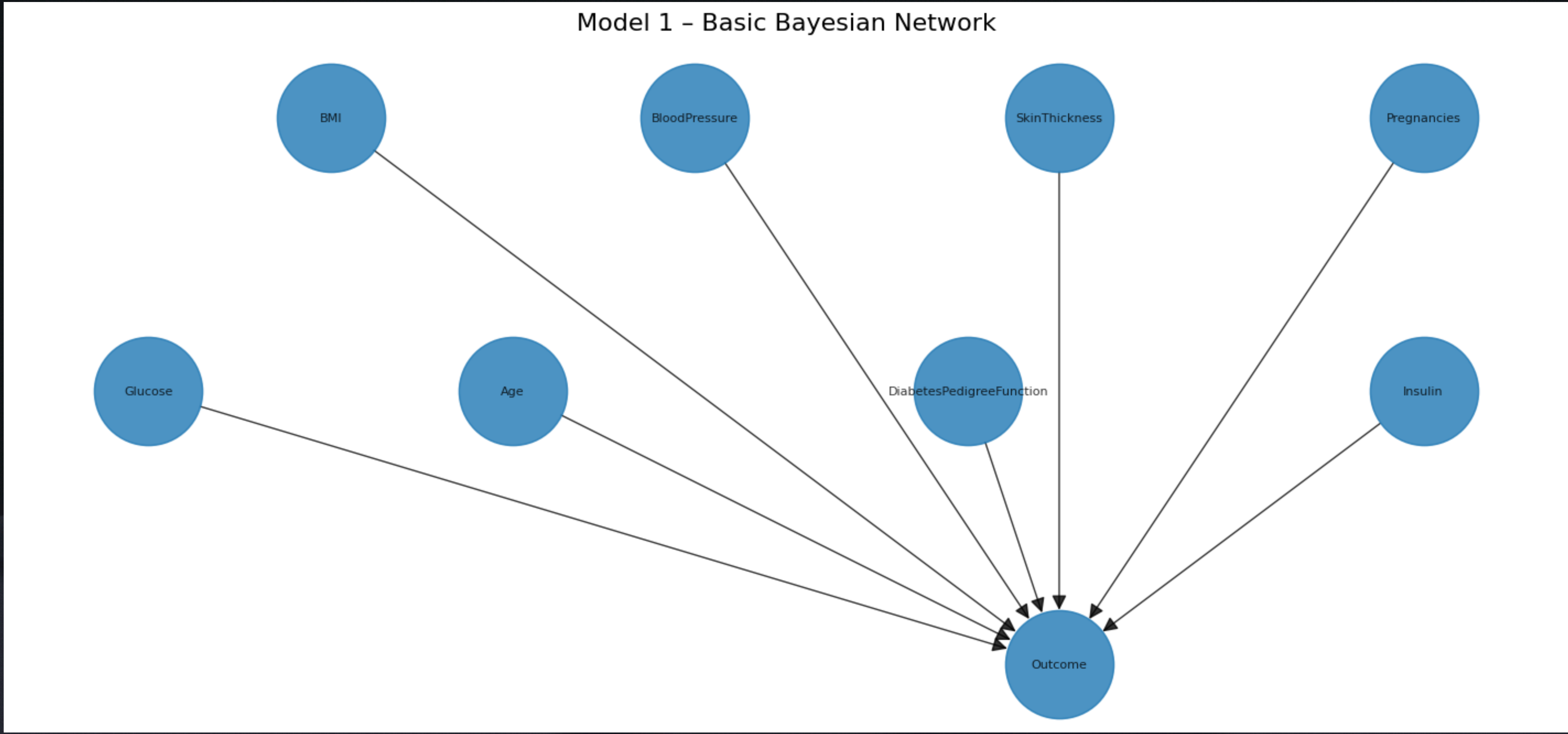

Pair Plot of Features vs Outcome
Pregnancies
Glucose
BloodPressure
SkinThickness
Insulin
BMI
DiabetesPedigreeFunction
Age
Outcome
0
1


Pair plot of each changed feature vs Outcome in KNN KMeans Imputed Data
BloodPressure
SkinThickness
BMI
DiabetesPedigreeFunction
Glucose
Outcome
0
1

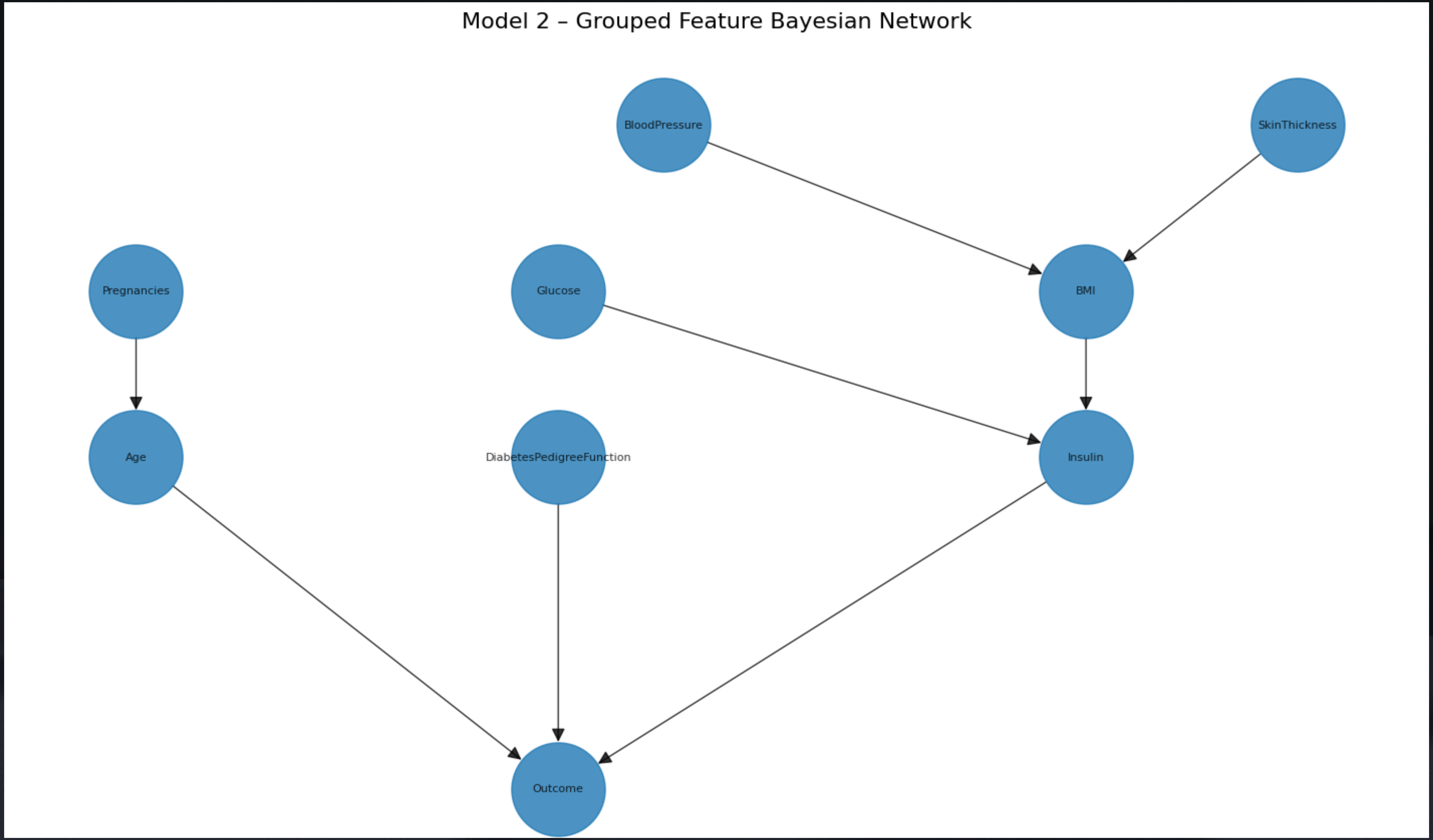
Training Bayesian Network

1. *Define the structure of the Bayesian Network which there is three options here:*
    - we can define the structure manually based on domain knowledge.Quantile Binning
    - we can use structure learning algorithms to learn the structure from data.
    - we can use a hybrid approach where we define some parts of the
2. *Find the Conditional Probability Distributions (CPDs) for each node in the network.*
3. *Perform inference on the Bayesian Network to make analyses with variable elimination. We will perform infrences to analyses the result of different scenarios.*

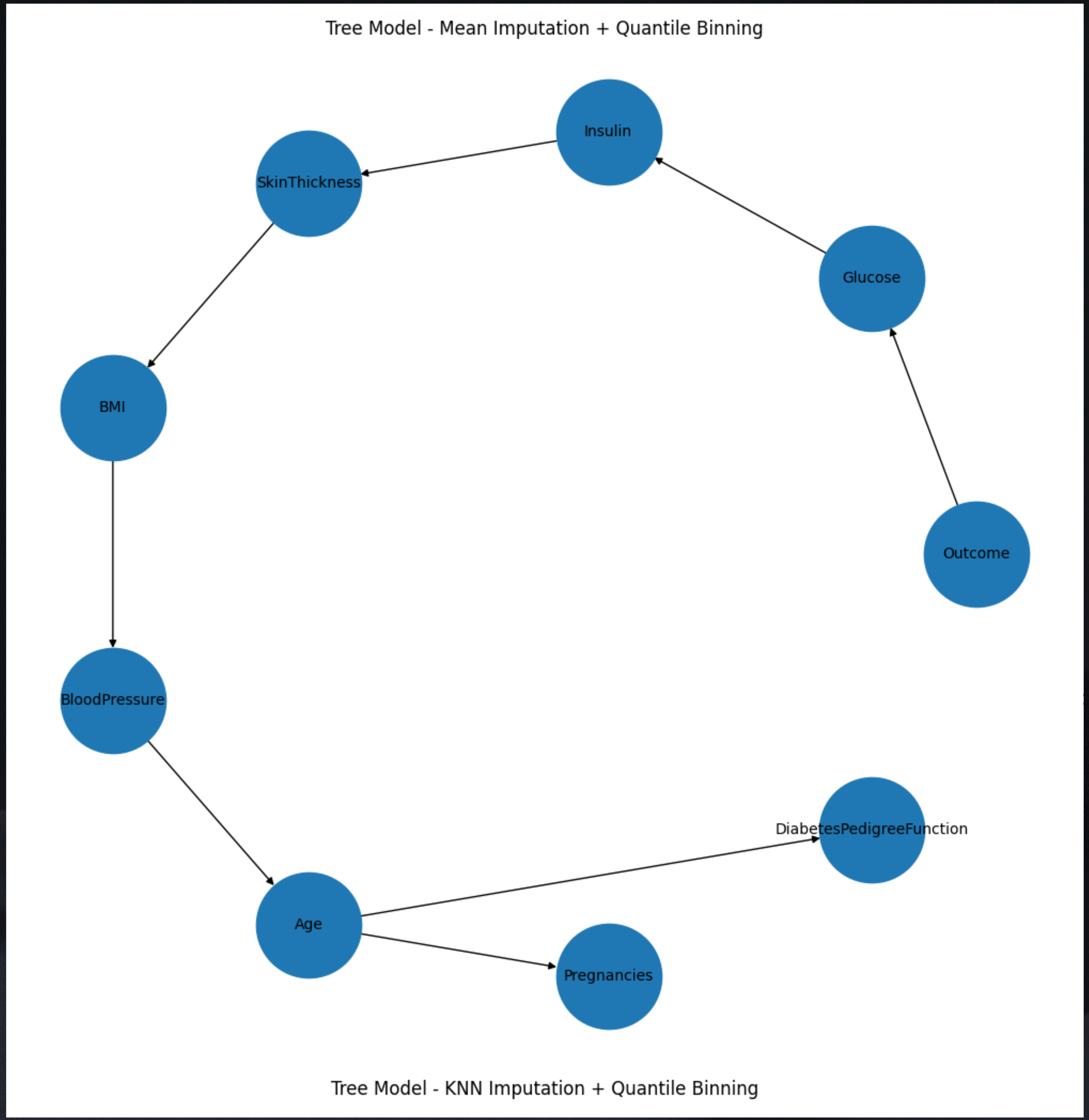
Basic Model

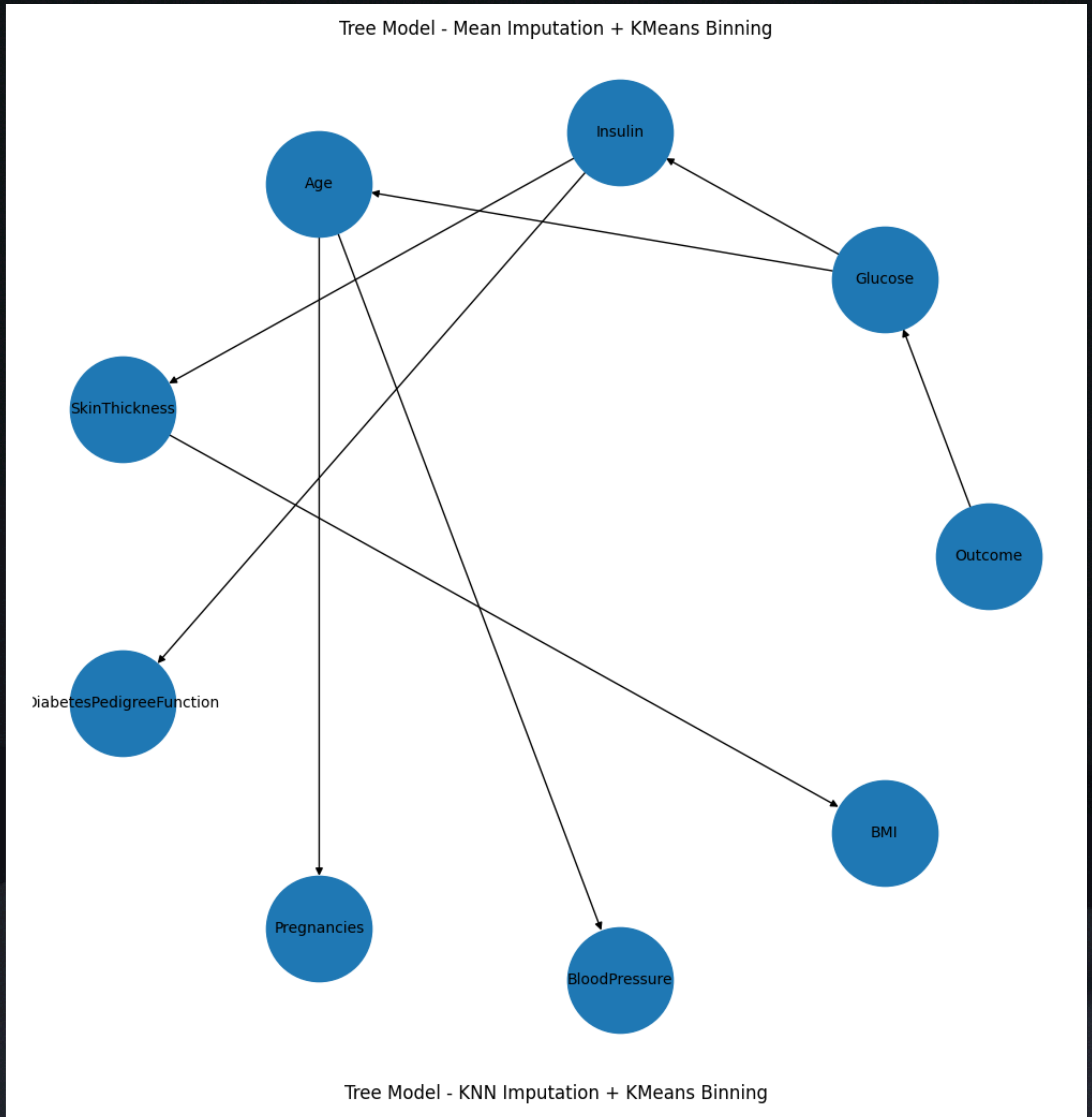

Model 1 – Basic Bayesian Network
BMI
BloodPressure
SkinThickness
Pregnancies
Glucose
Age
DiabetesPedigreeFunction
Insulin
Outcome

Grouped Feature Model

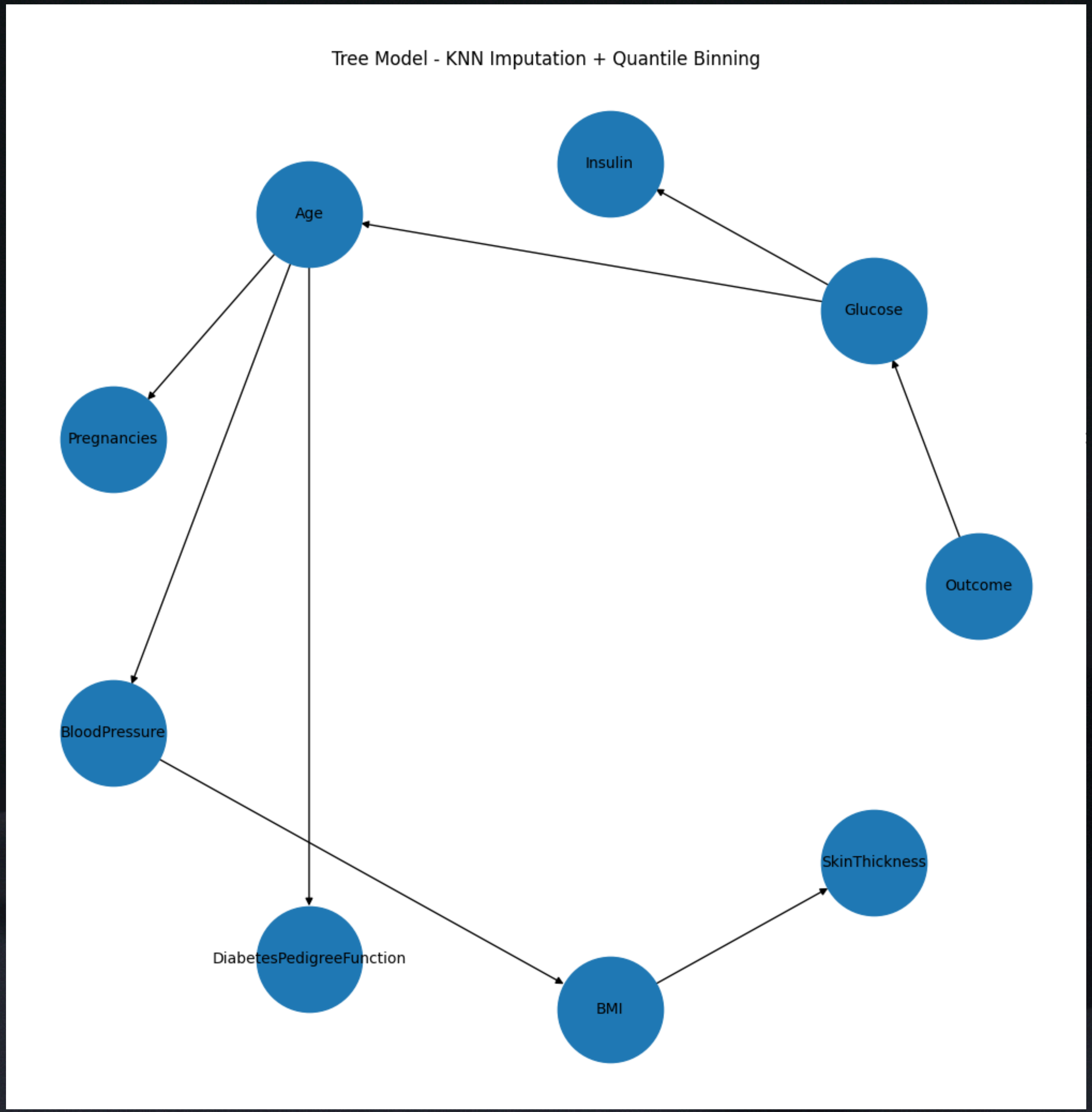


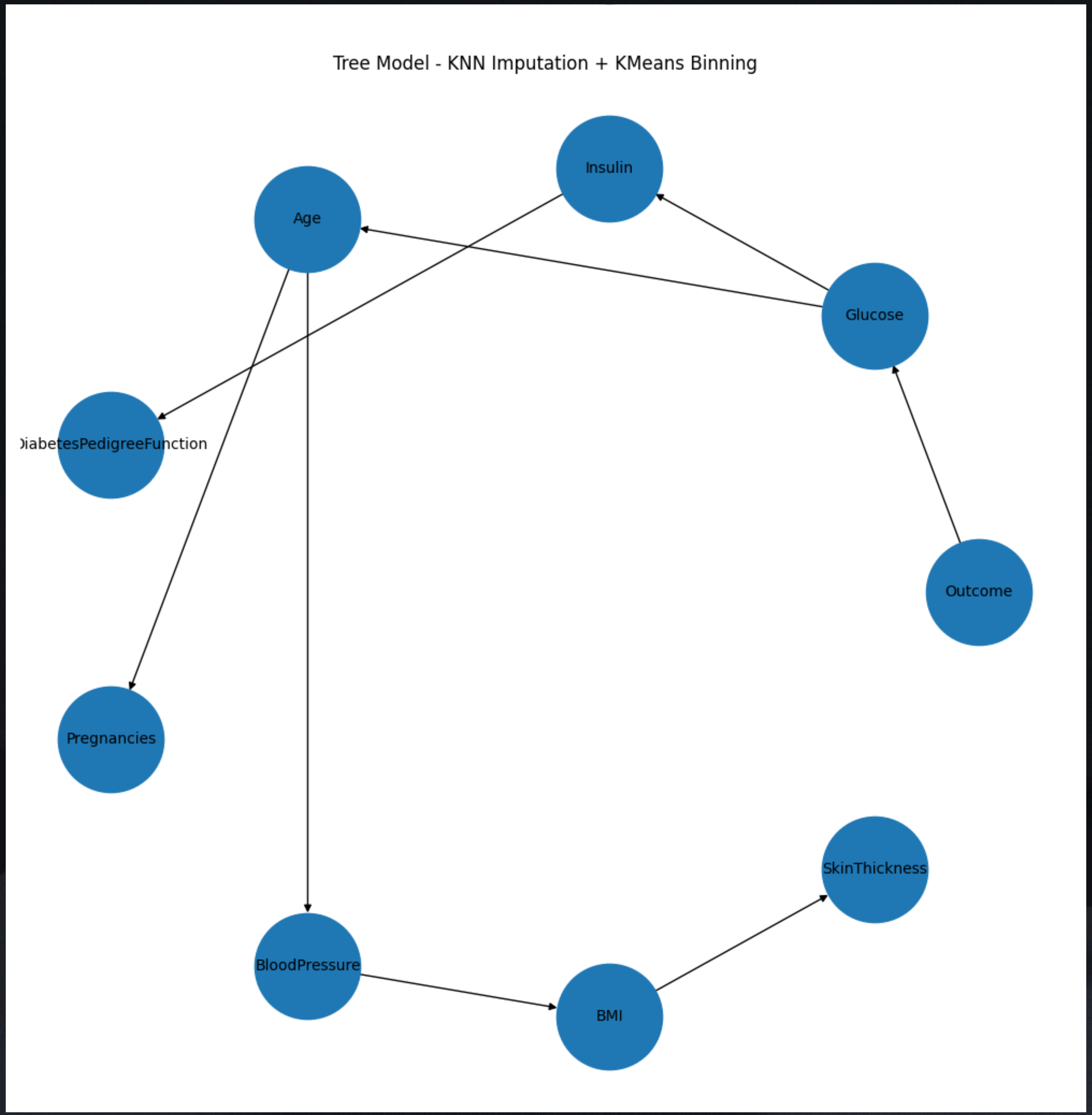
Tree Models


Tree Model - Mean Imputation + Quantile Binning
Insulin
SkinThickness
Glucose
BMI
Outcome
BloodPressure
DiabetesPedigreeFunction
Age
Pregnancies
Tree Model - KNN Imputation + Quantile Binning

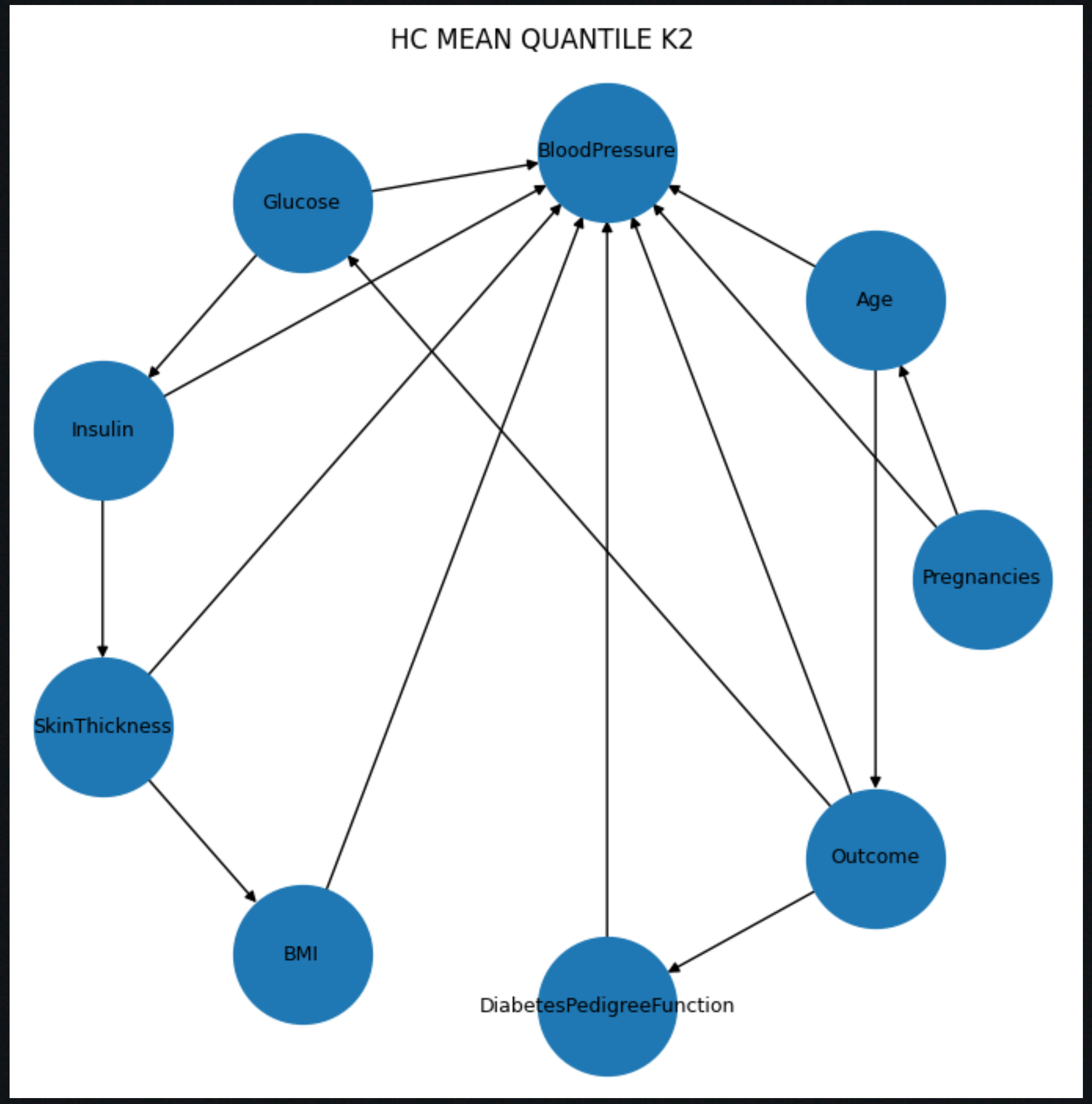

Tree Model - Mean Imputation + KMeans Binning
Insulin
Age
Glucose
SkinThickness
Outcome
DiabetesPedigreeFunction
BMI
Pregnancies
BloodPressure
Tree Model - KNN Imputation + KMeans Binning

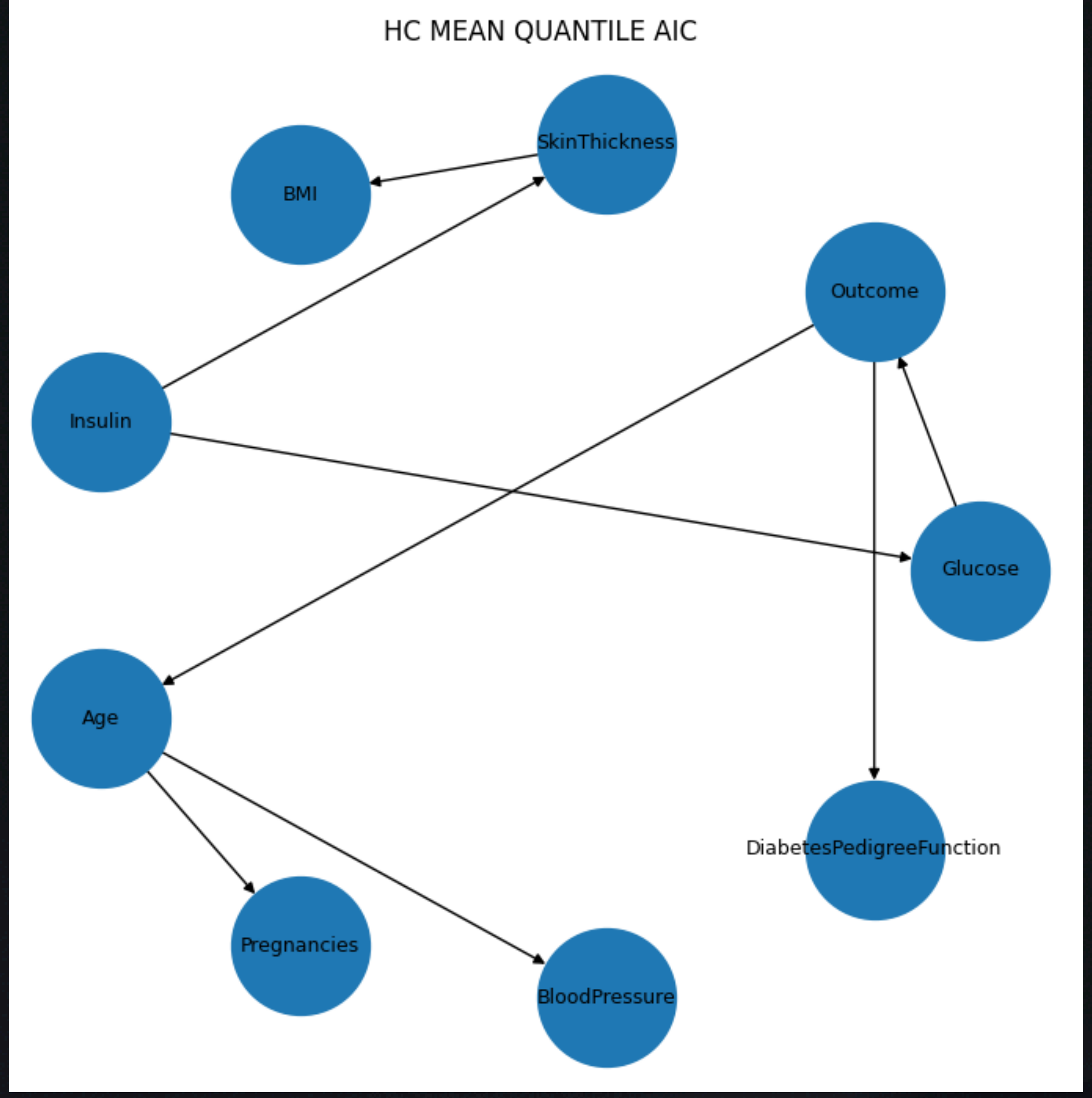
Tree Models

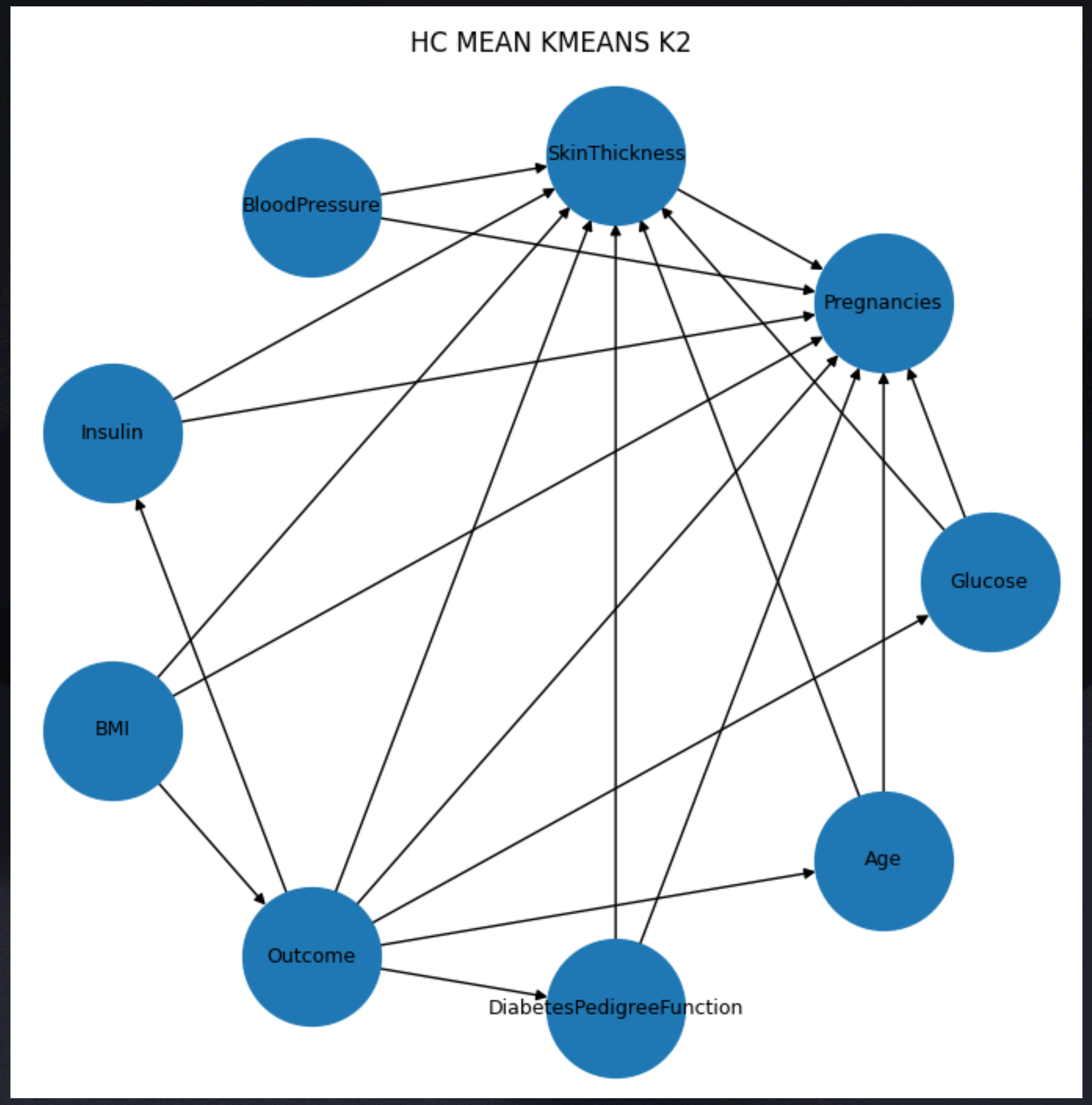

Tree Model - KNN Imputation + Quantile Binning
Insulin
Age
Glucose
Pregnancies
Outcome
BloodPressure
SkinThickness
DiabetesPedigreeFunction
BMI

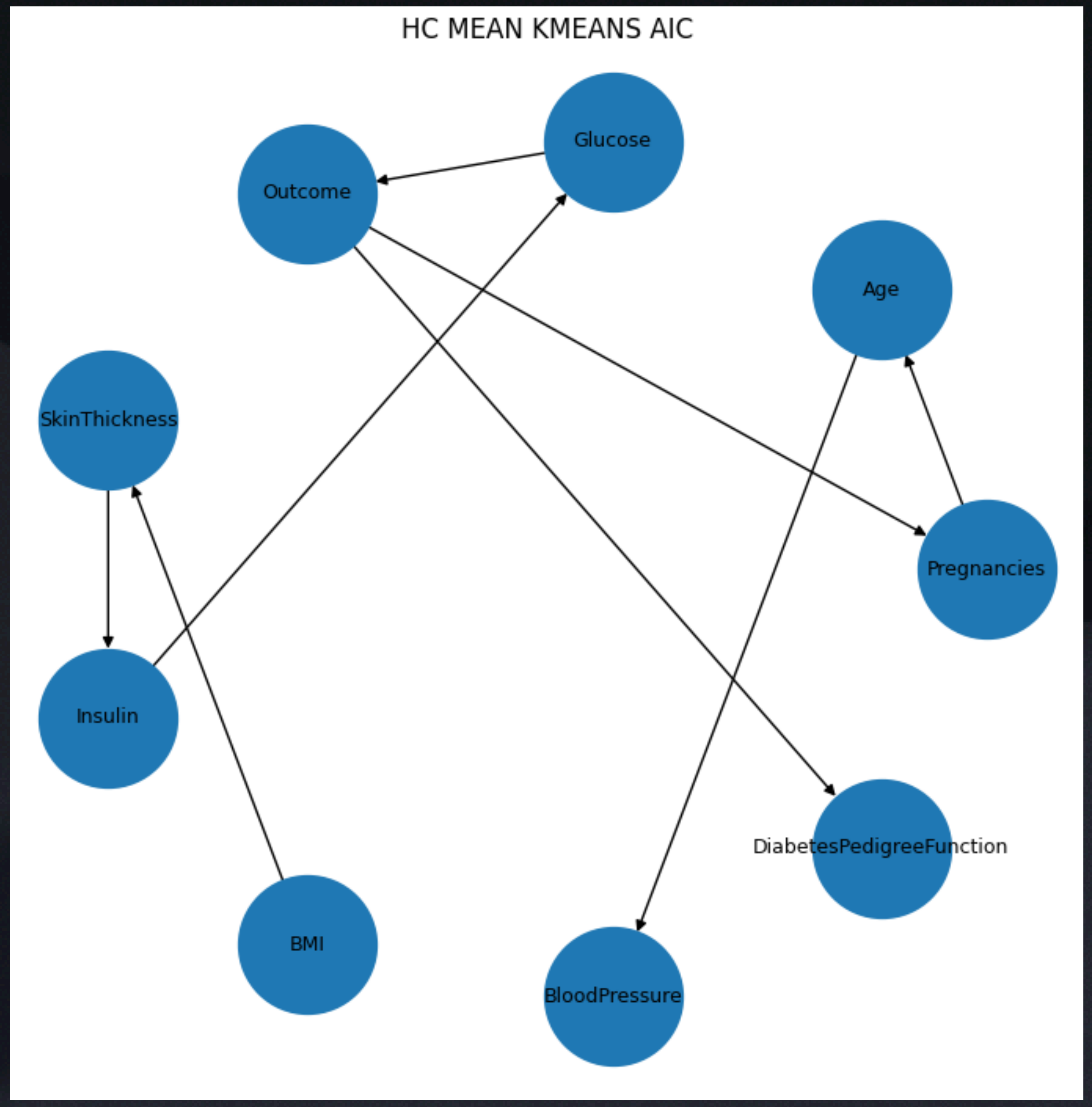

Tree Model - KNN Imputation + KMeans Binning
Insulin
Age
Glucose
DiabetesPedigreeFunction
Outcome
Pregnancies
SkinThickness
BloodPressure
BMI

Hill Climbing

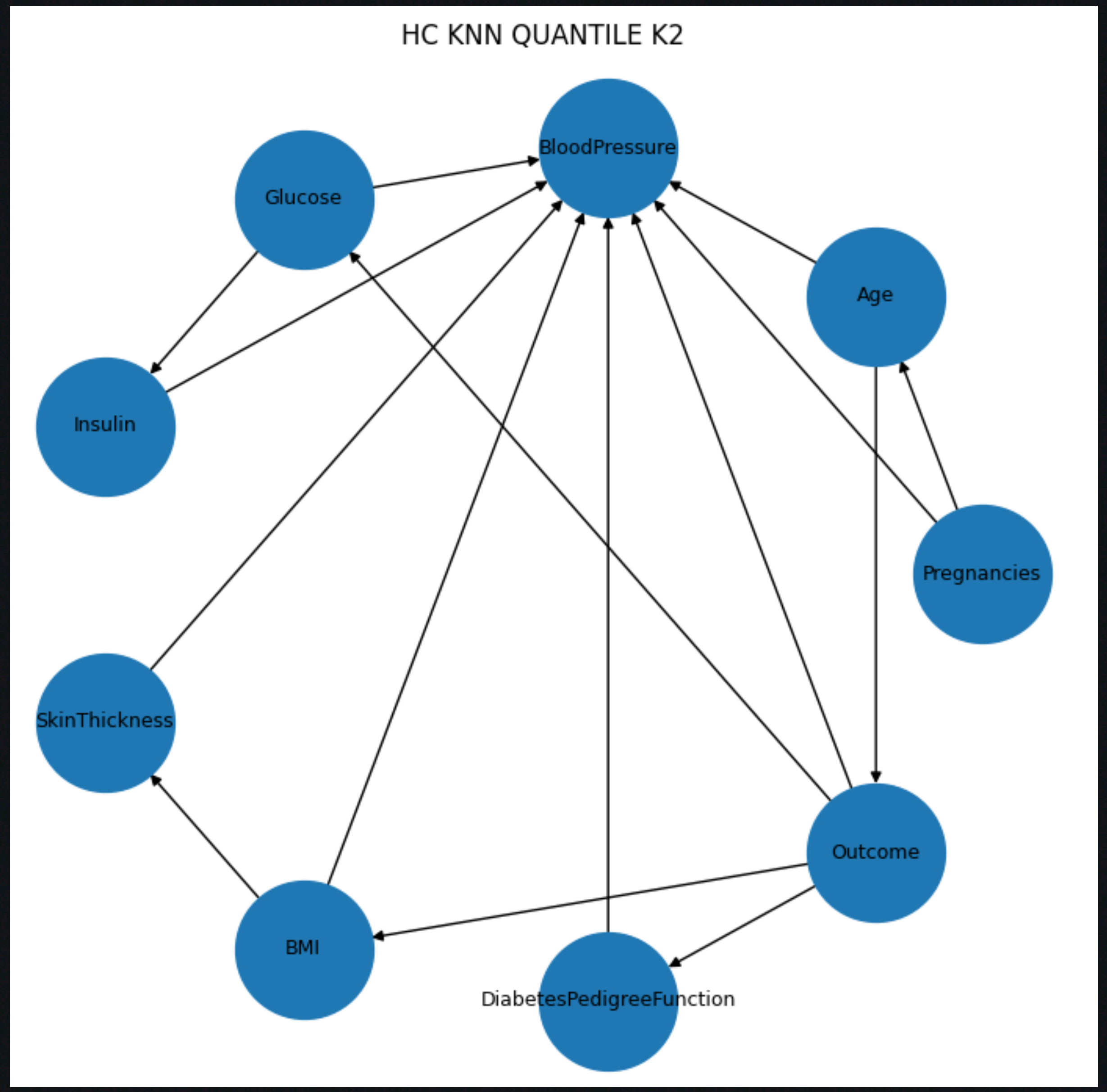


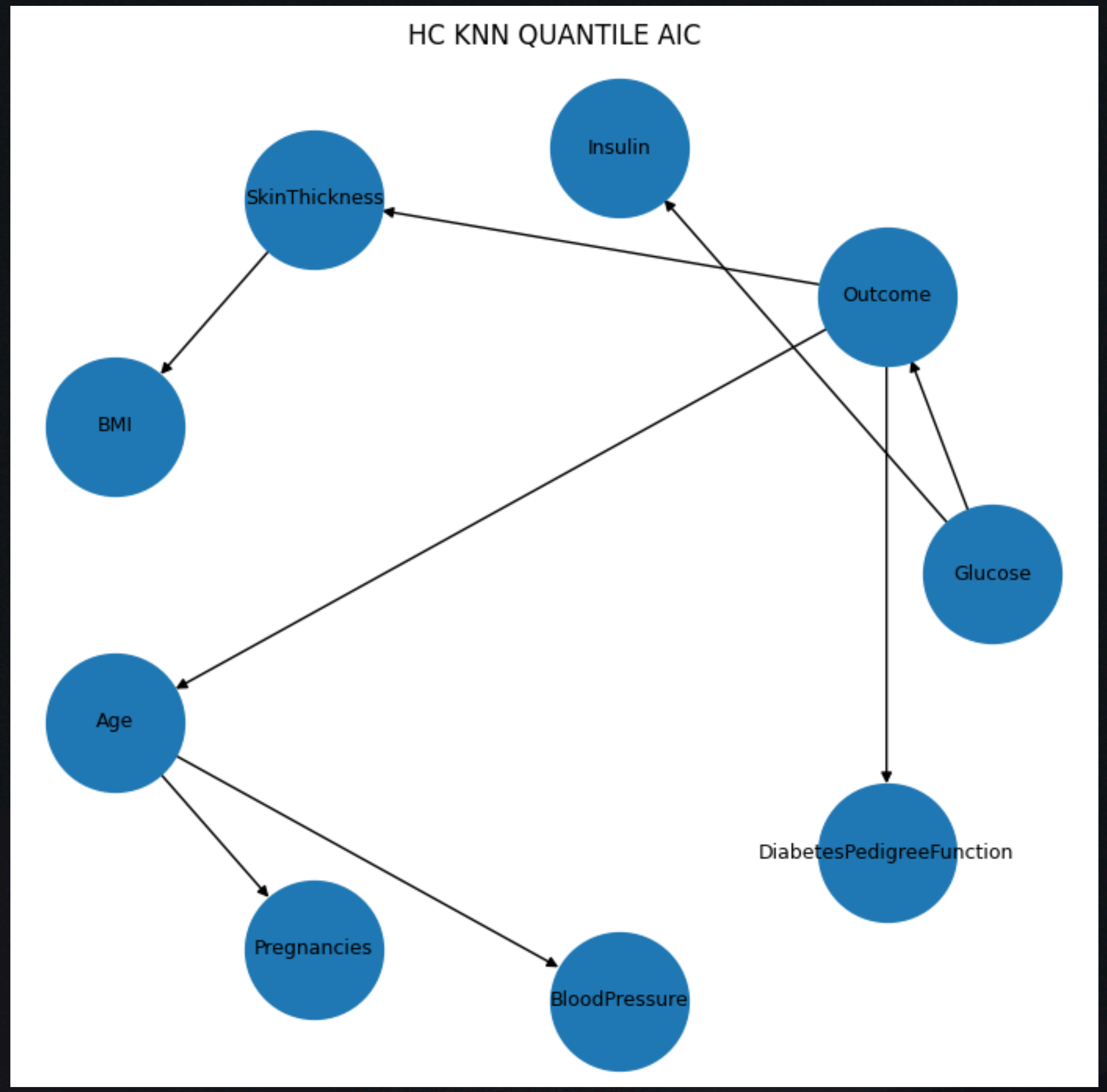


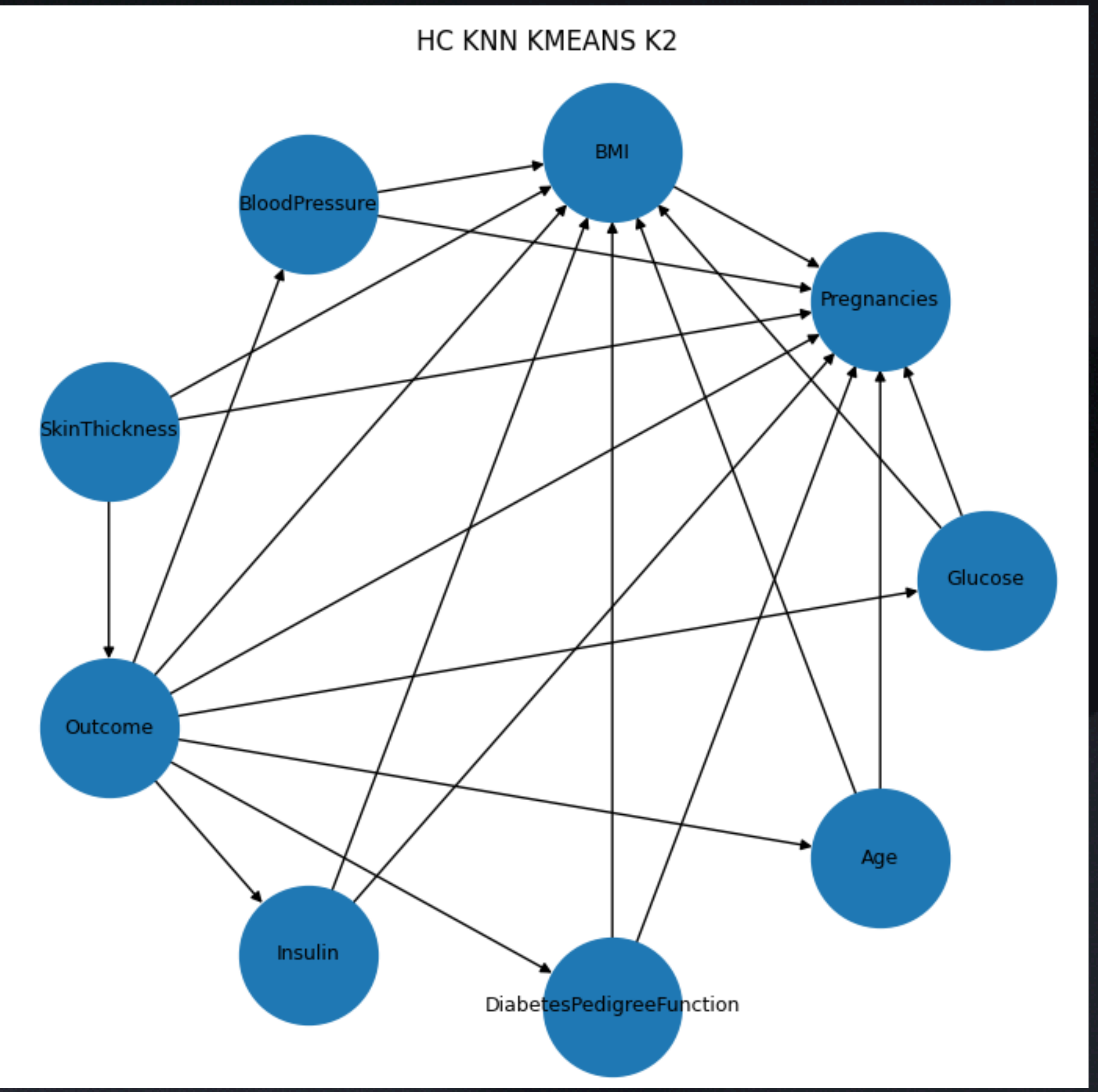


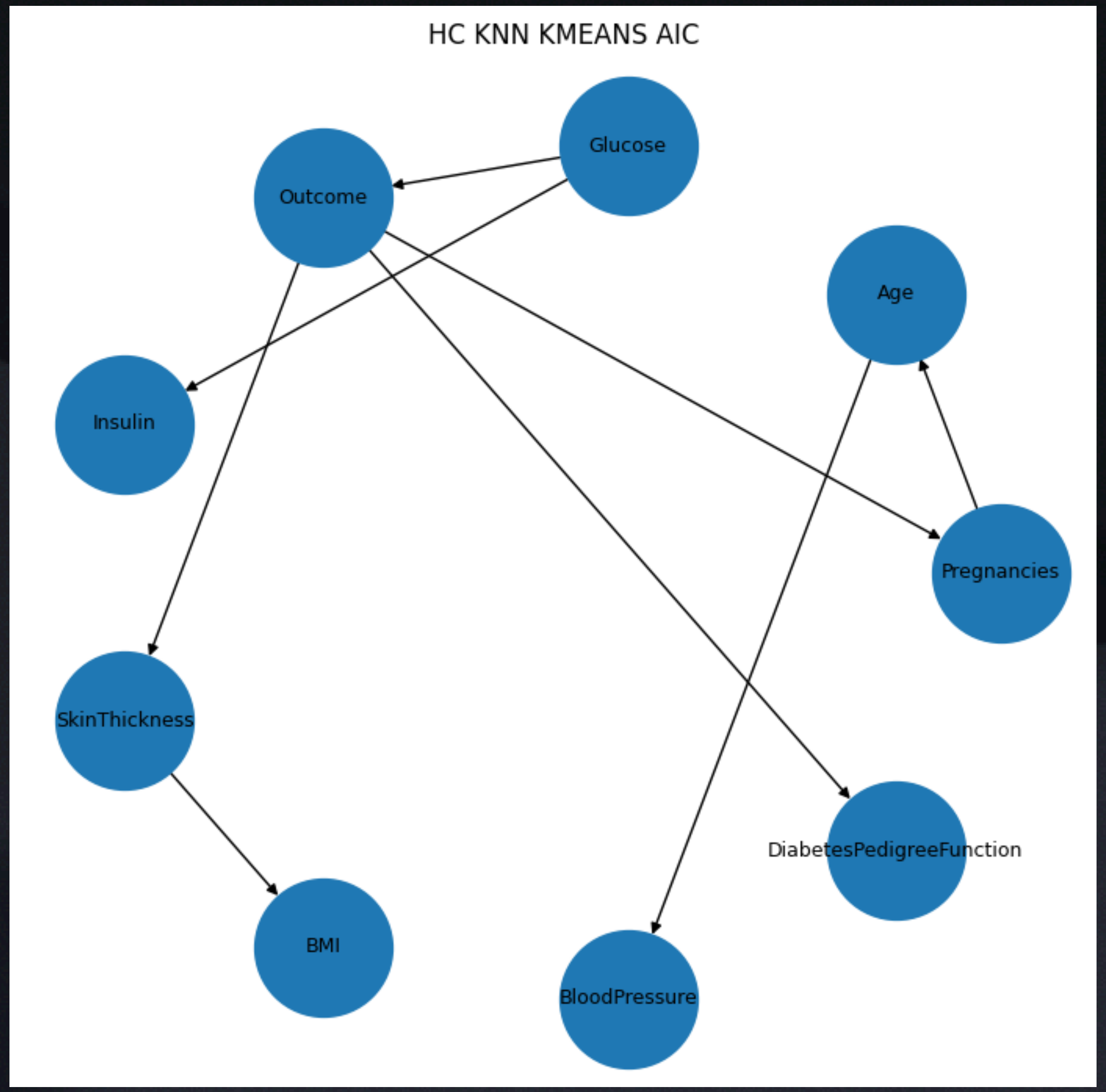


Hill Climbing









Training

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Analyzing model: hc_knn_kmeans_aic
The total number of valid independence assertions is 28
Local semantics of the current model:
(Pregnancies ⊥ BMI, SkinThickness, DiabetesPedigreeFunction, Glucose, Insulin | Outcome)
(BloodPressure ⊥ BMI, SkinThickness, Outcome, Pregnancies, DiabetesPedigreeFunction, Glucose, Insulin | Age)
(SkinThickness ⊥ Age, Pregnancies, DiabetesPedigreeFunction, BloodPressure, Glucose, Insulin | Outcome)
(Insulin ⊥ BMI, Age, Outcome, Pregnancies, BloodPressure, DiabetesPedigreeFunction, SkinThickness | Glucose)
(BMI ⊥ Age, Outcome, Pregnancies, BloodPressure, DiabetesPedigreeFunction, Glucose, Insulin | SkinThickness)
(DiabetesPedigreeFunction ⊥ BMI, Age, SkinThickness, Pregnancies, BloodPressure, Glucose, Insulin | Outcome)
(Age ⊥ BMI, SkinThickness, Outcome, DiabetesPedigreeFunction, Glucose, Insulin | Pregnancies)
(Outcome ⊥ Insulin | Glucose)
Checking Markov blankets
The Markov blanket of node Pregnancies is ['Age', 'Outcome']
The Markov blanket of node Glucose is ['Outcome', 'Insulin']
The Markov blanket of node BloodPressure is ['Age']
The Markov blanket of node SkinThickness is ['BMI', 'Outcome']
The Markov blanket of node Insulin is ['Glucose']
The Markov blanket of node BMI is ['SkinThickness']
The Markov blanket of node DiabetesPedigreeFunction is ['Outcome']
The Markov blanket of node Age is ['BloodPressure', 'Pregnancies']
The Markov blanket of node Outcome is ['Pregnancies', 'DiabetesPedigreeFunction', 'SkinThickness', 'Glucose']
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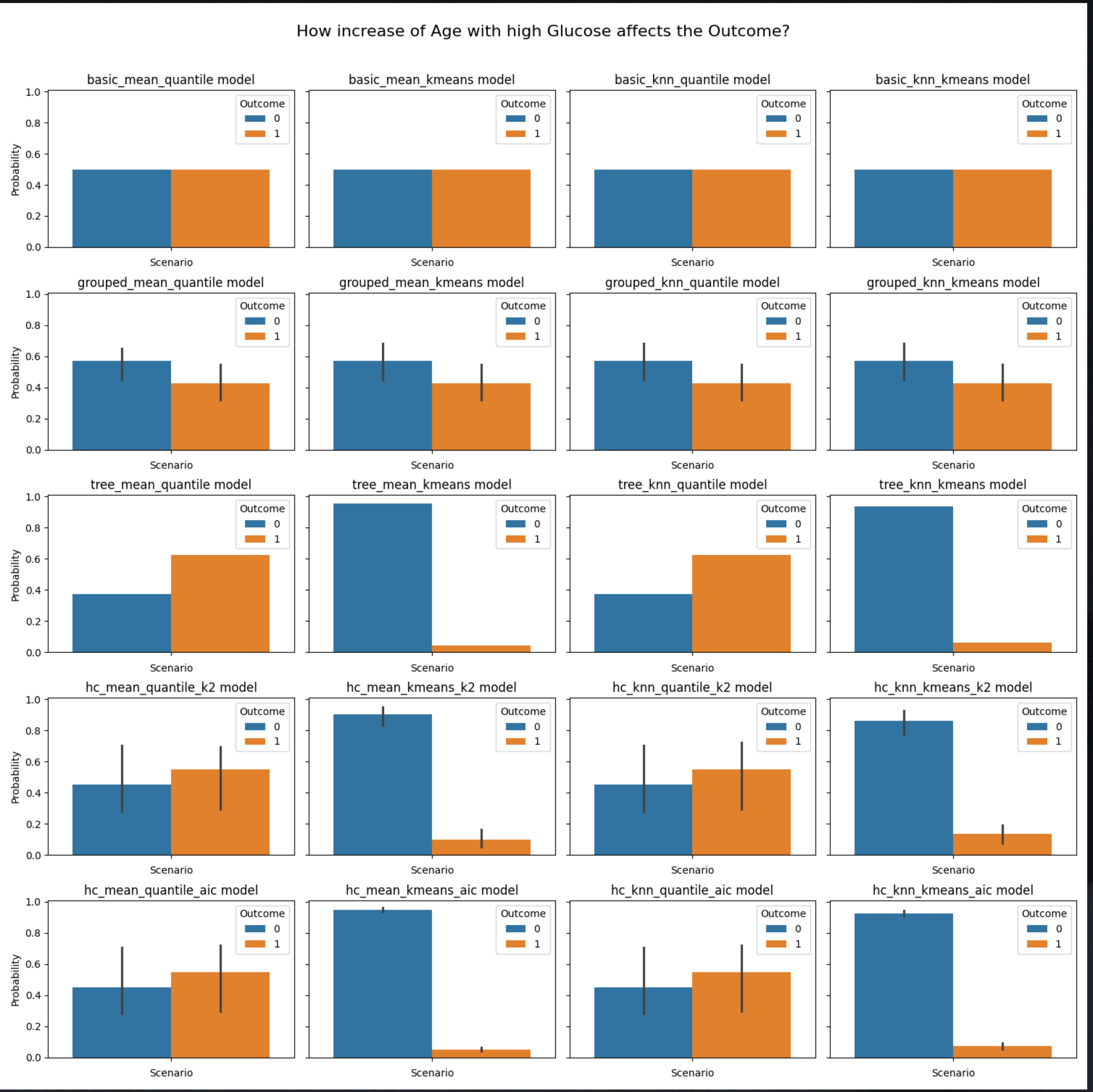
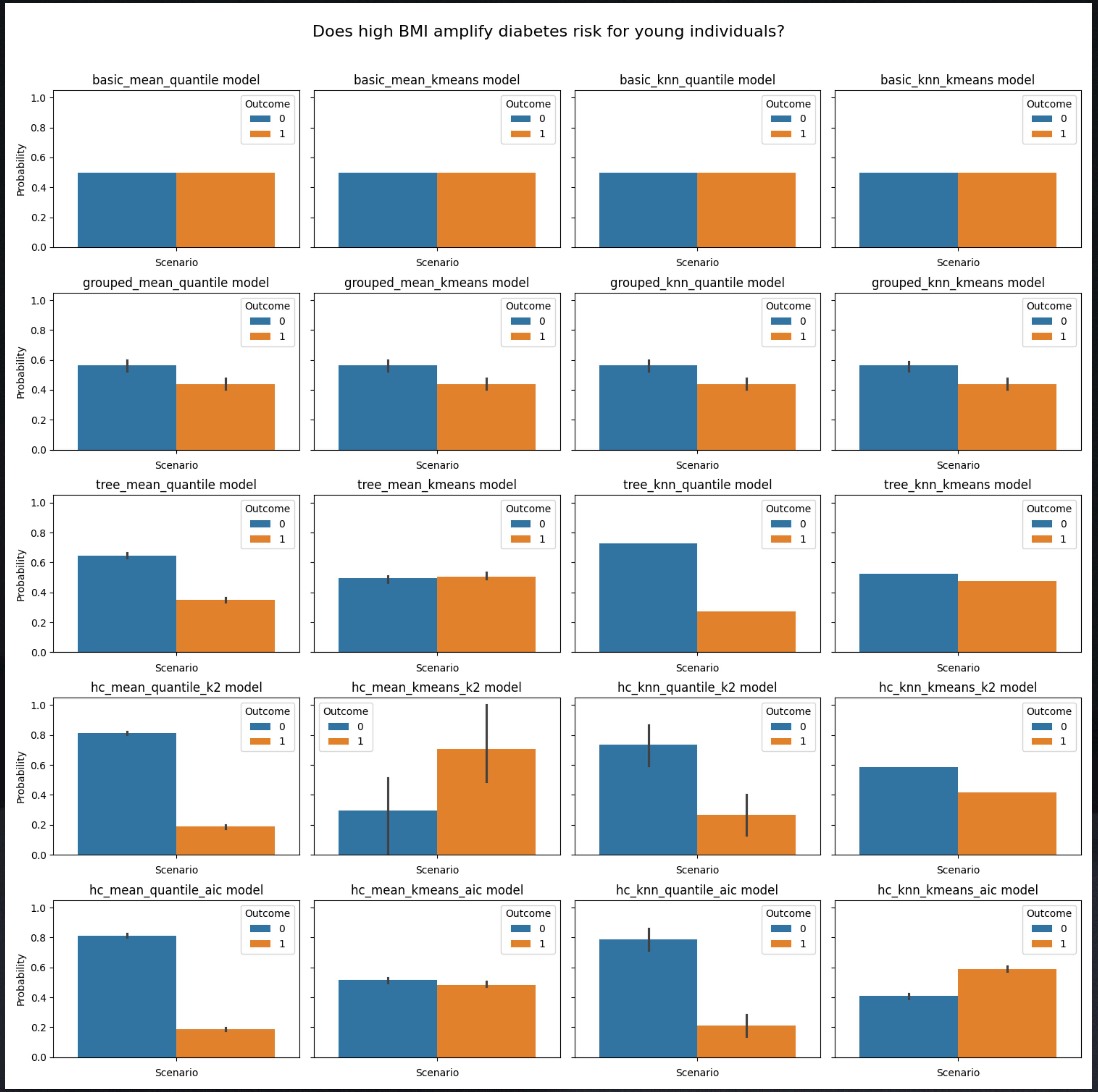
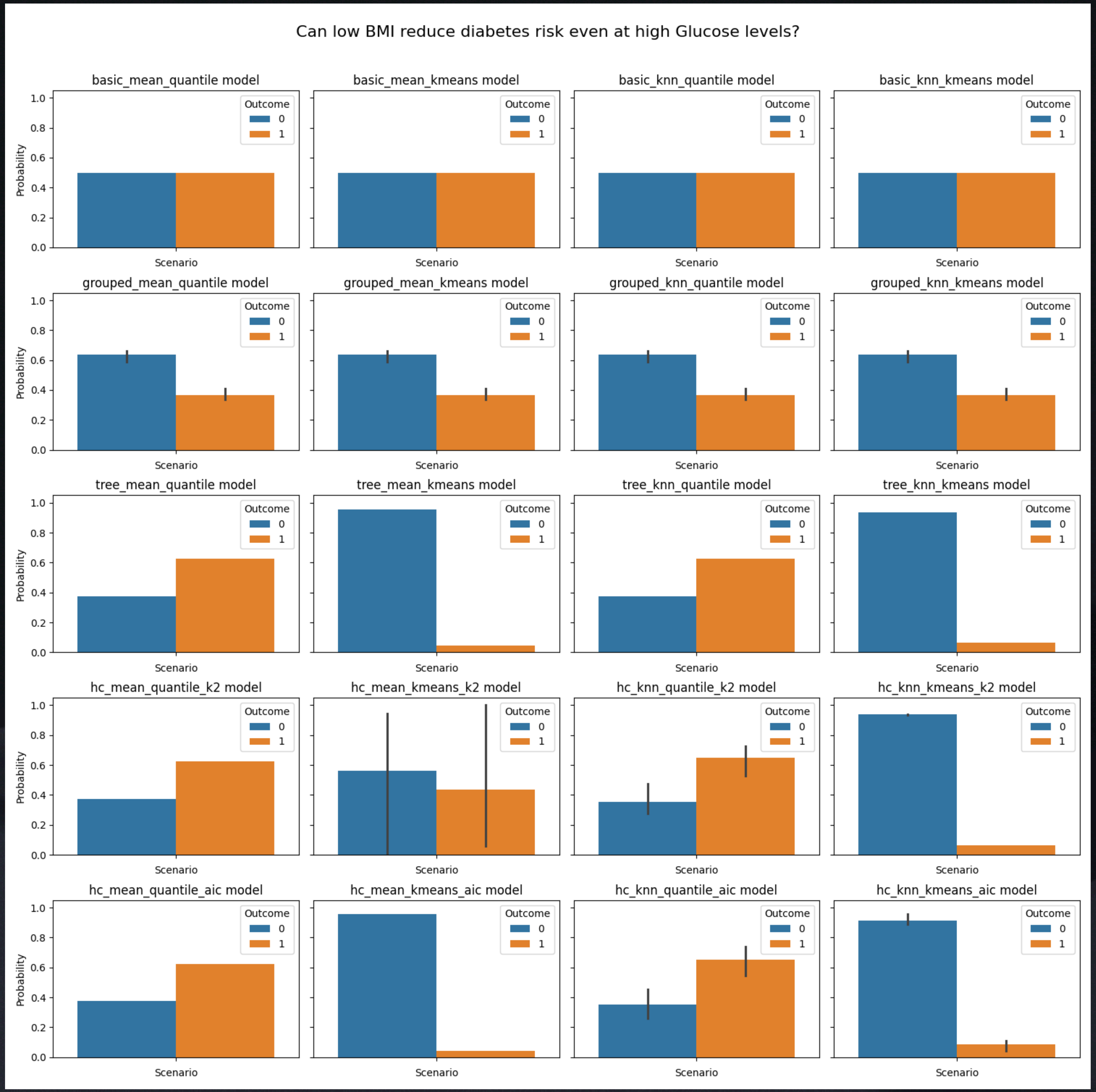
Question 1: How does diabetes risk change when Age increases while Glucose remains high?

Question 2: Does high BMI amplify diabetes risk for young individuals?

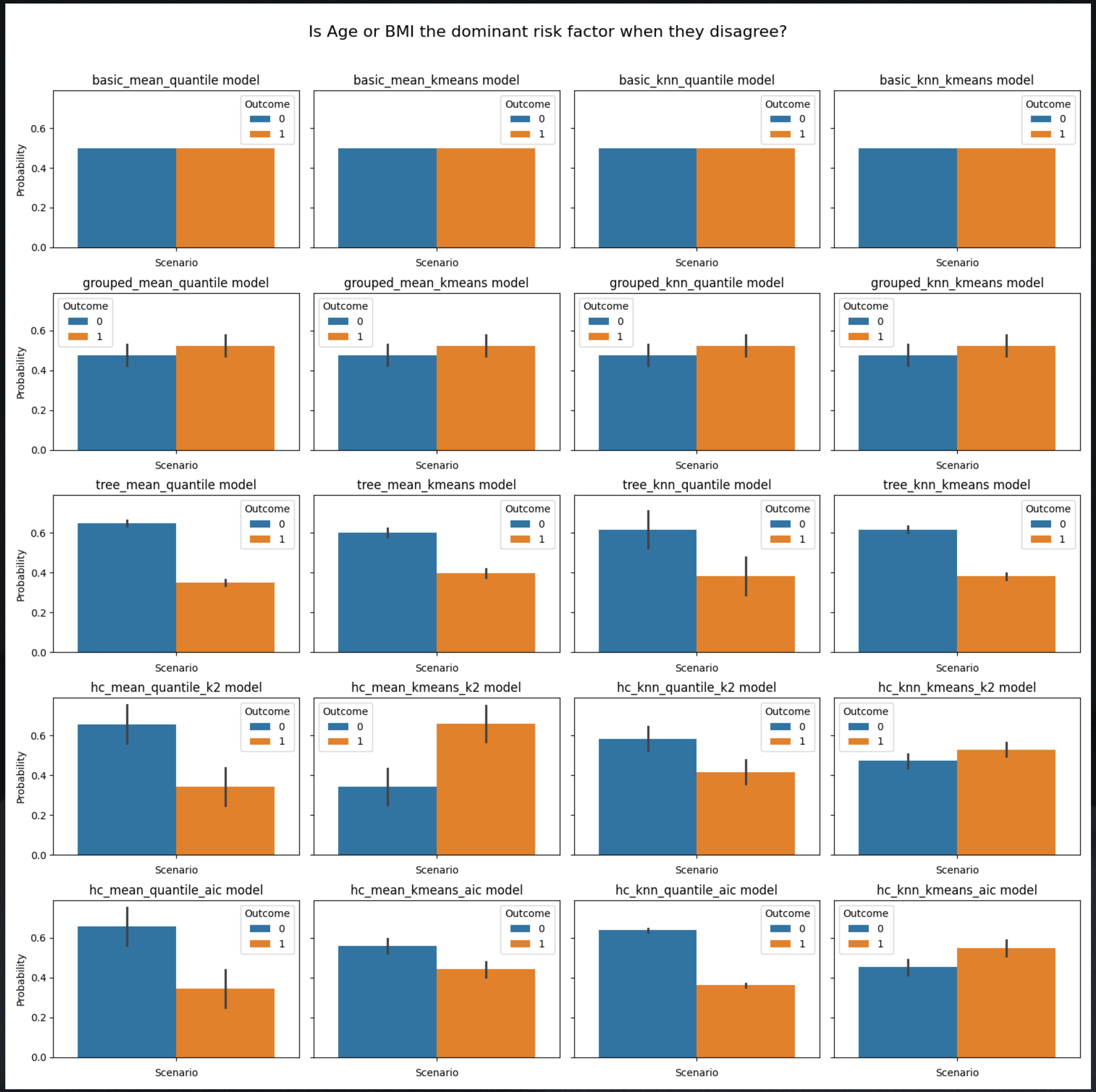




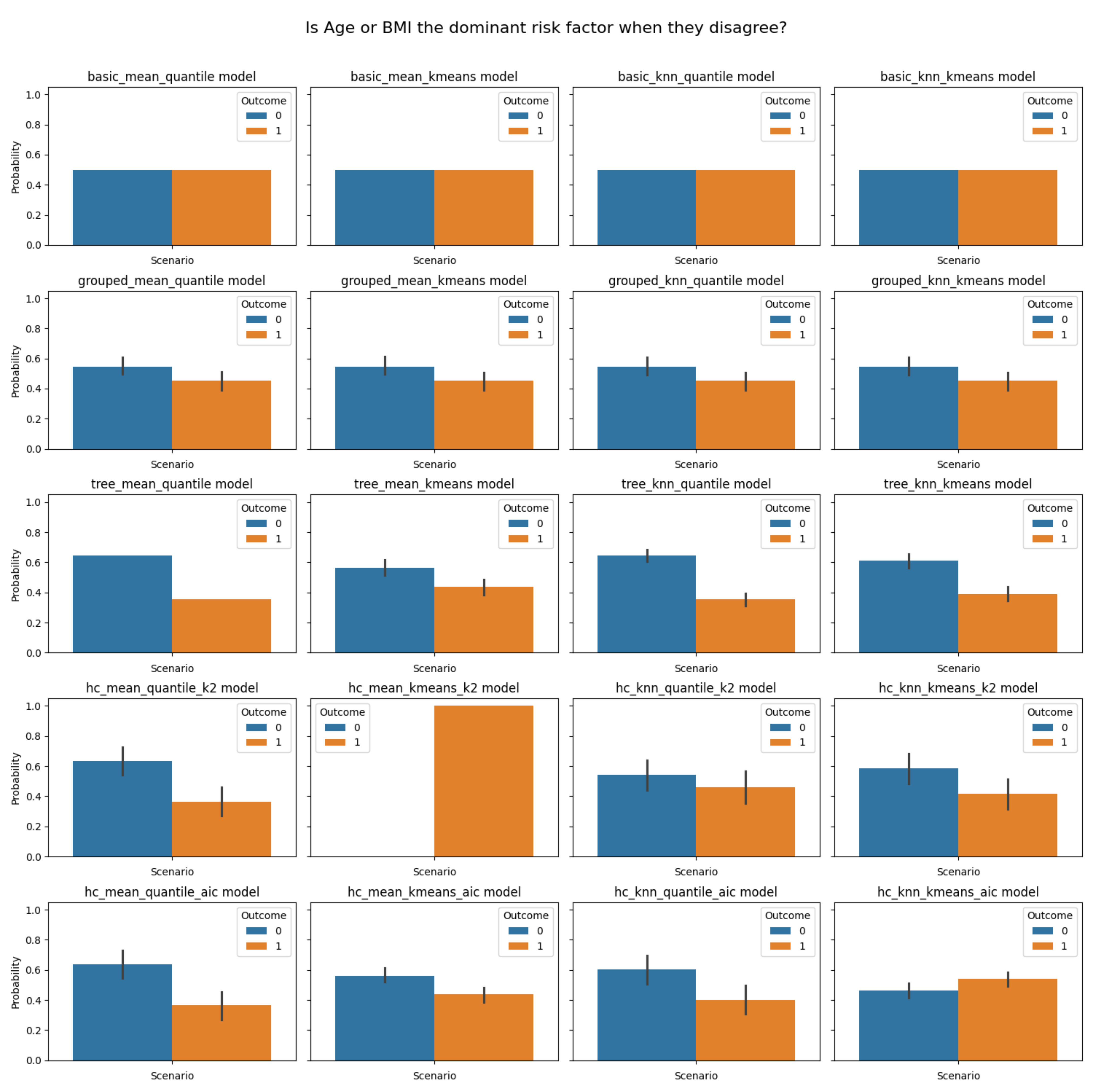
Question 3: Can low BMI reduce diabetes risk even at high Glucose levels?



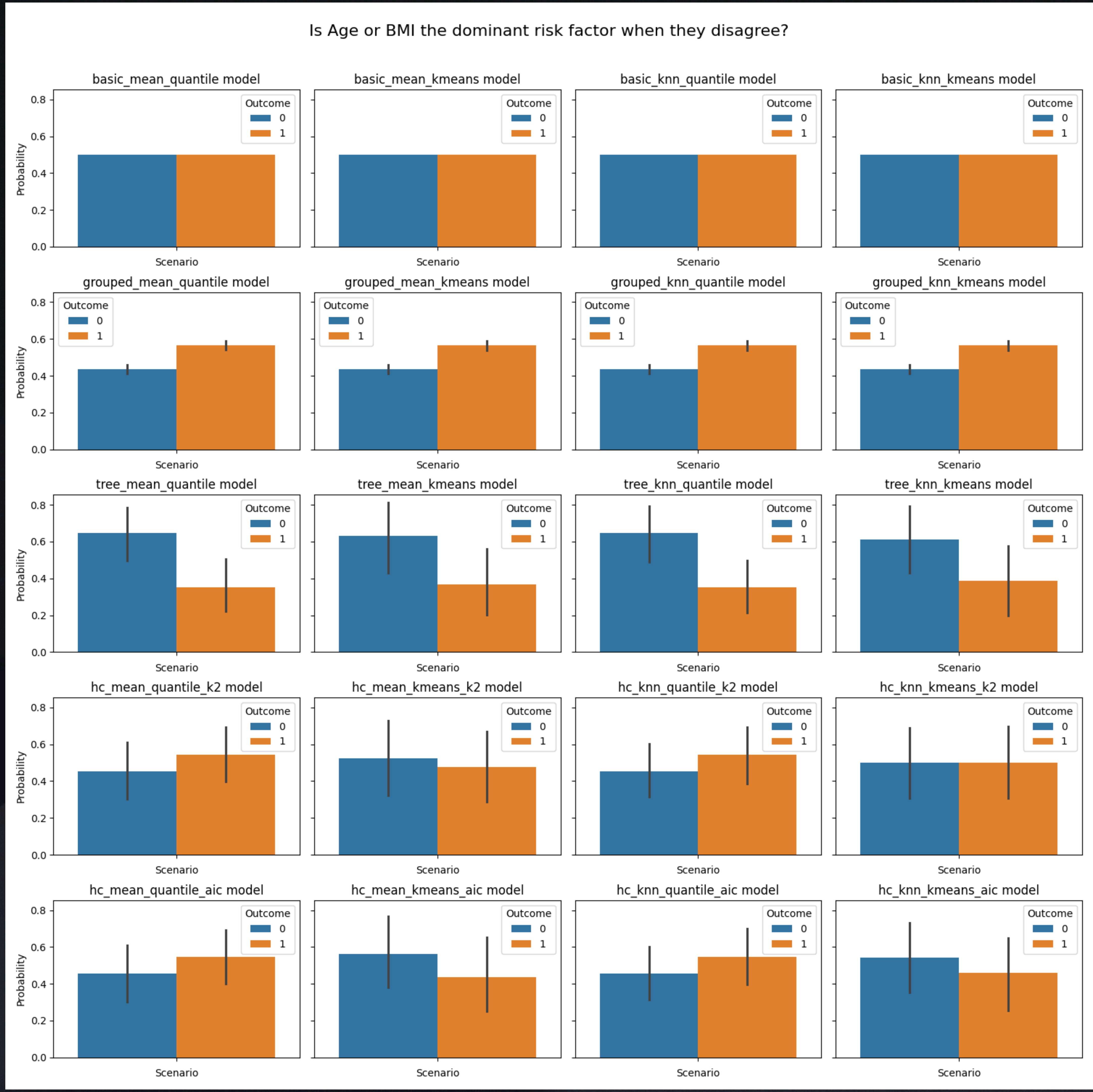
Question 4: Is Age or BMI the dominant risk factor when they disagree?



Question 5: How does diabetes probability evolve across Age bins for average BMI?

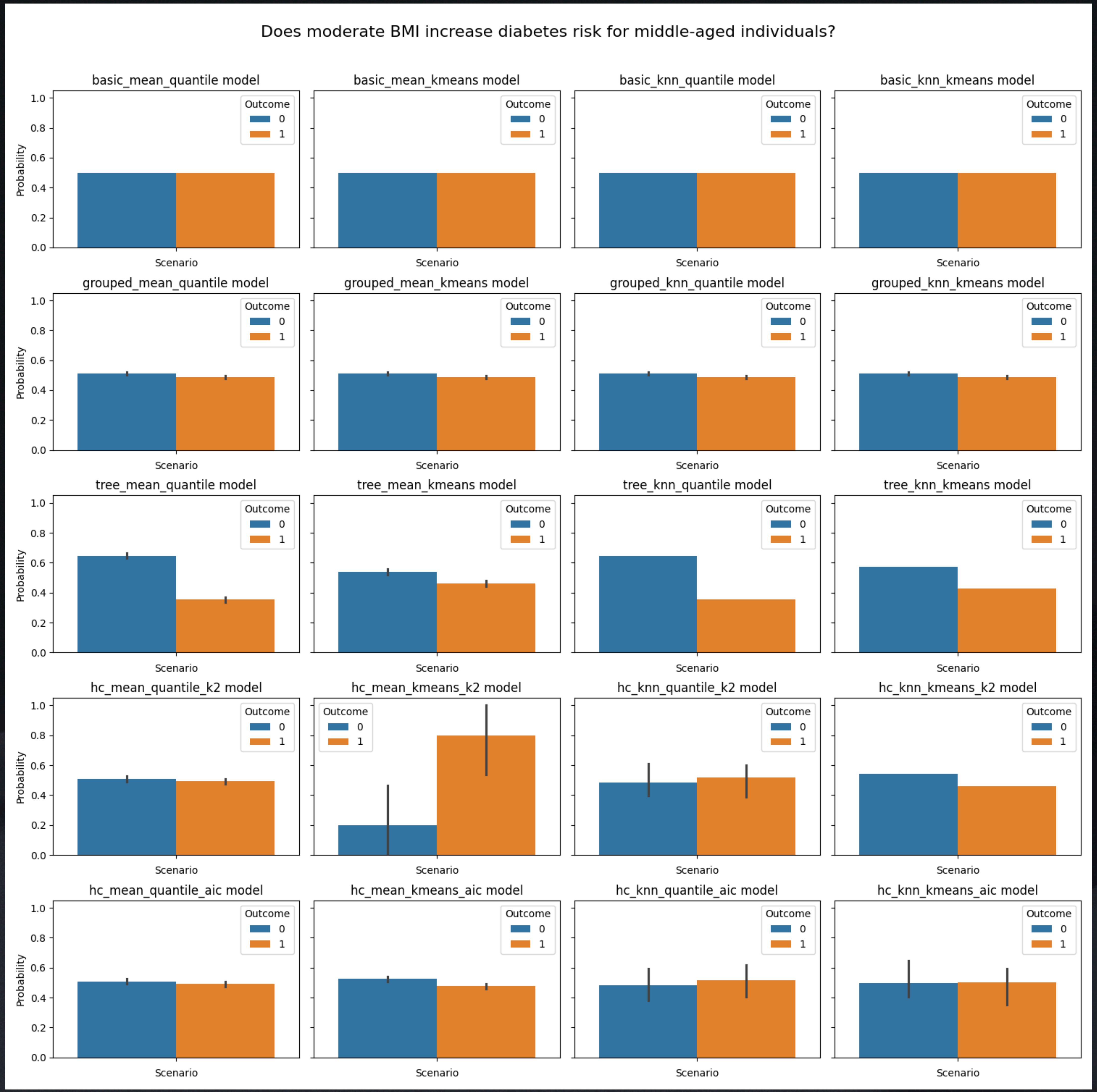


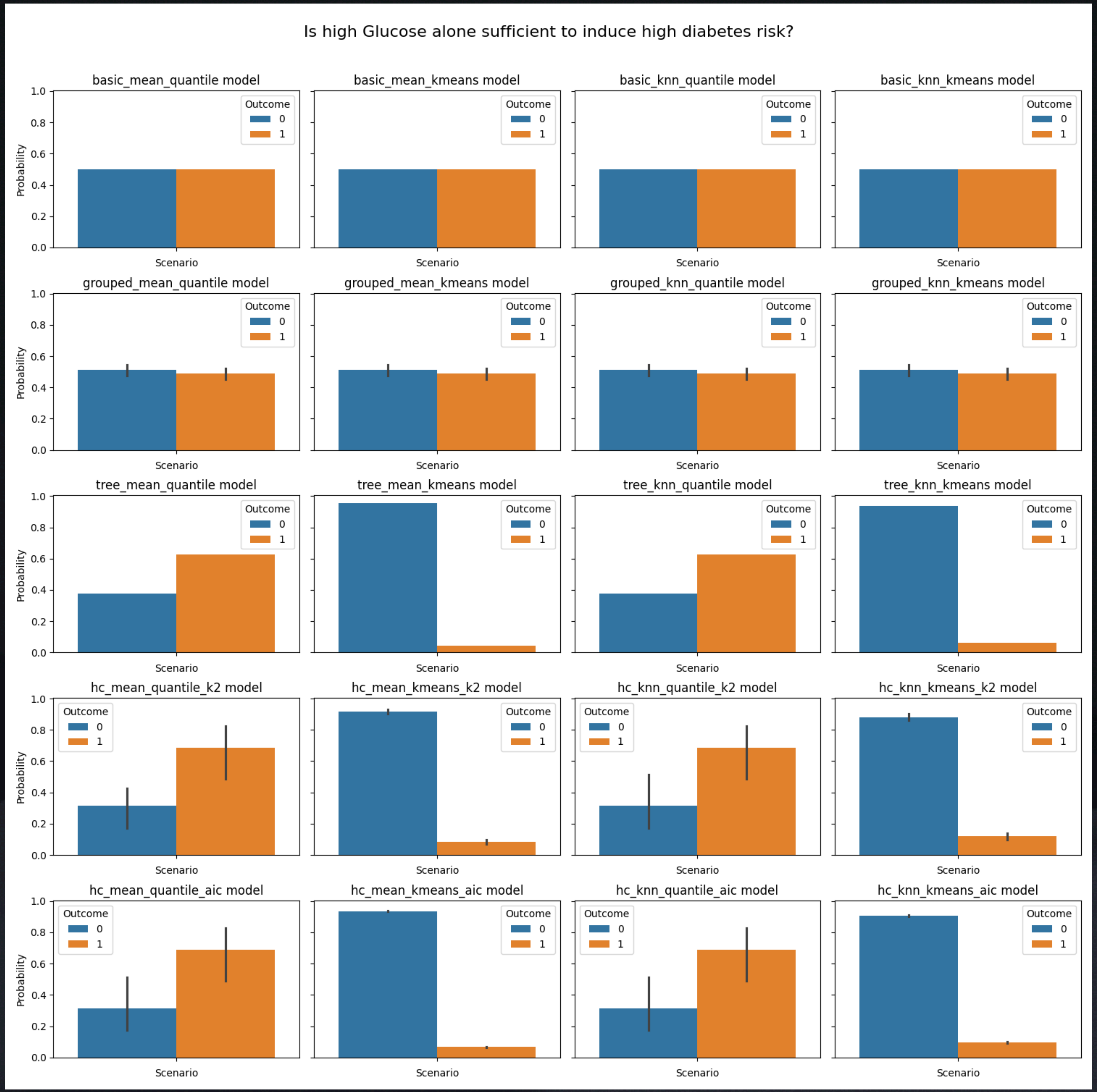
Question 6: At what Glucose level does diabetes risk exceed 50% for elderly individuals?



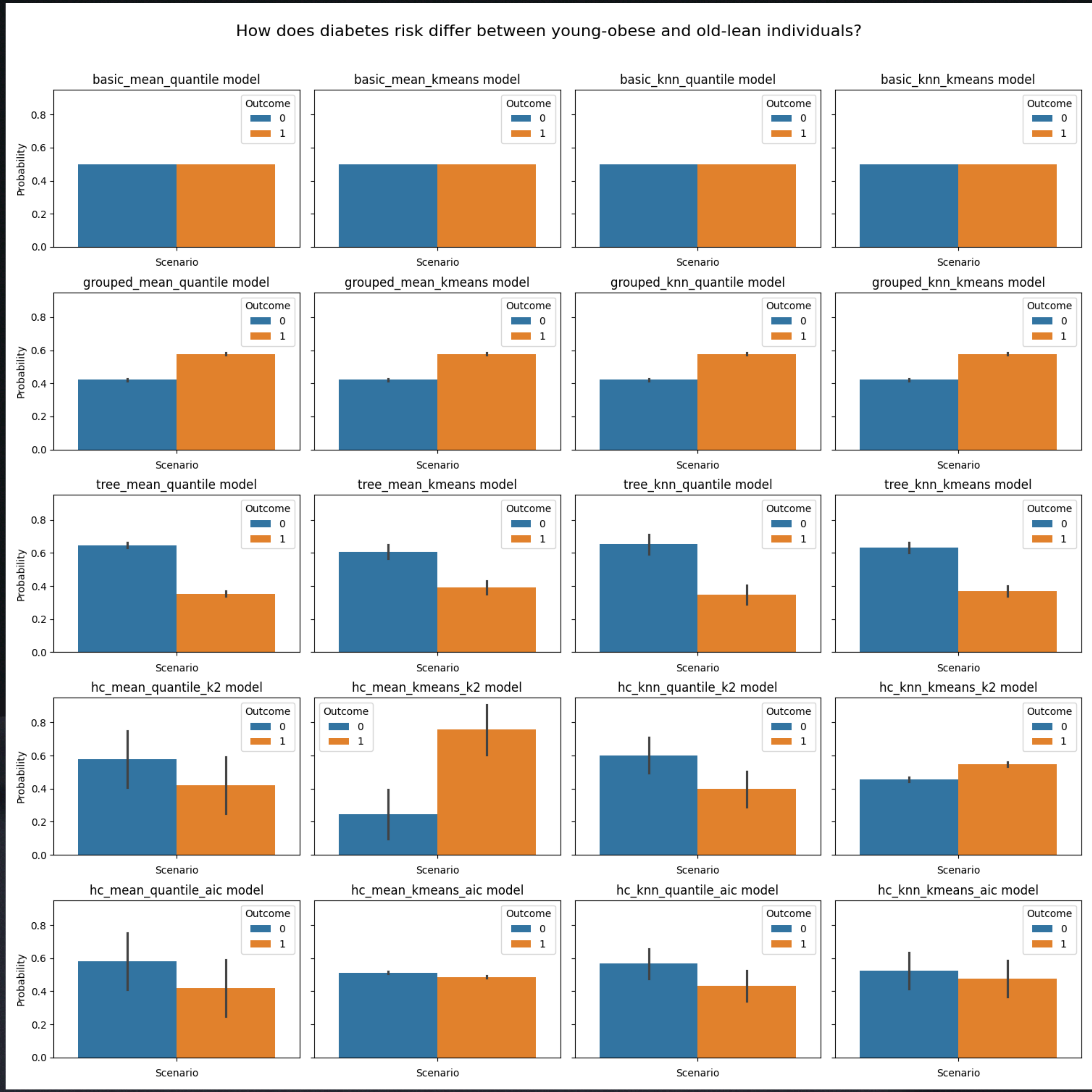
Question 7: Does moderate BMI increase diabetes risk for middle-aged individuals?

Question 8: Is high Glucose alone sufficient to induce high diabetes risk?

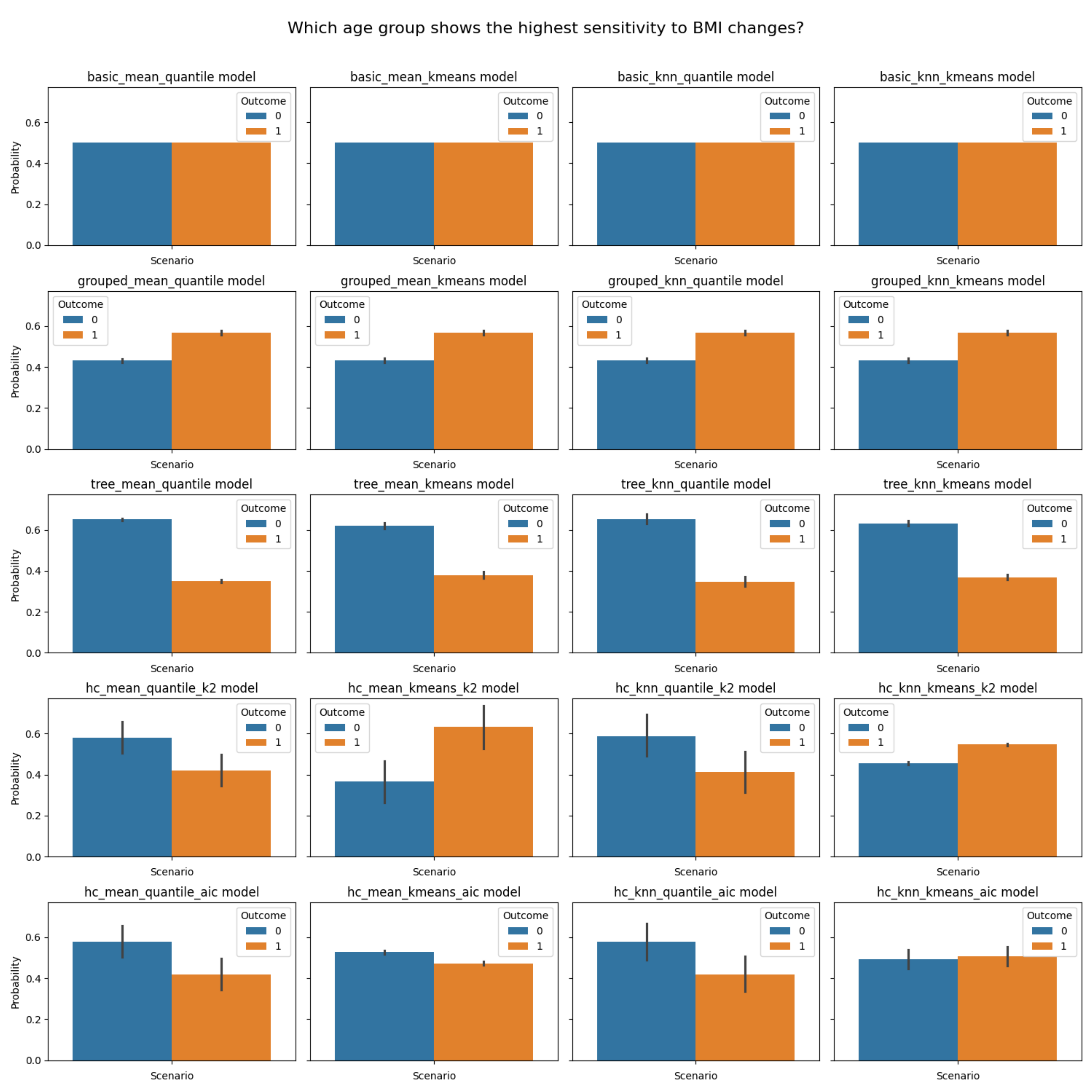




Question 9: How does diabetes risk differ between young-obese and old-lean individuals?

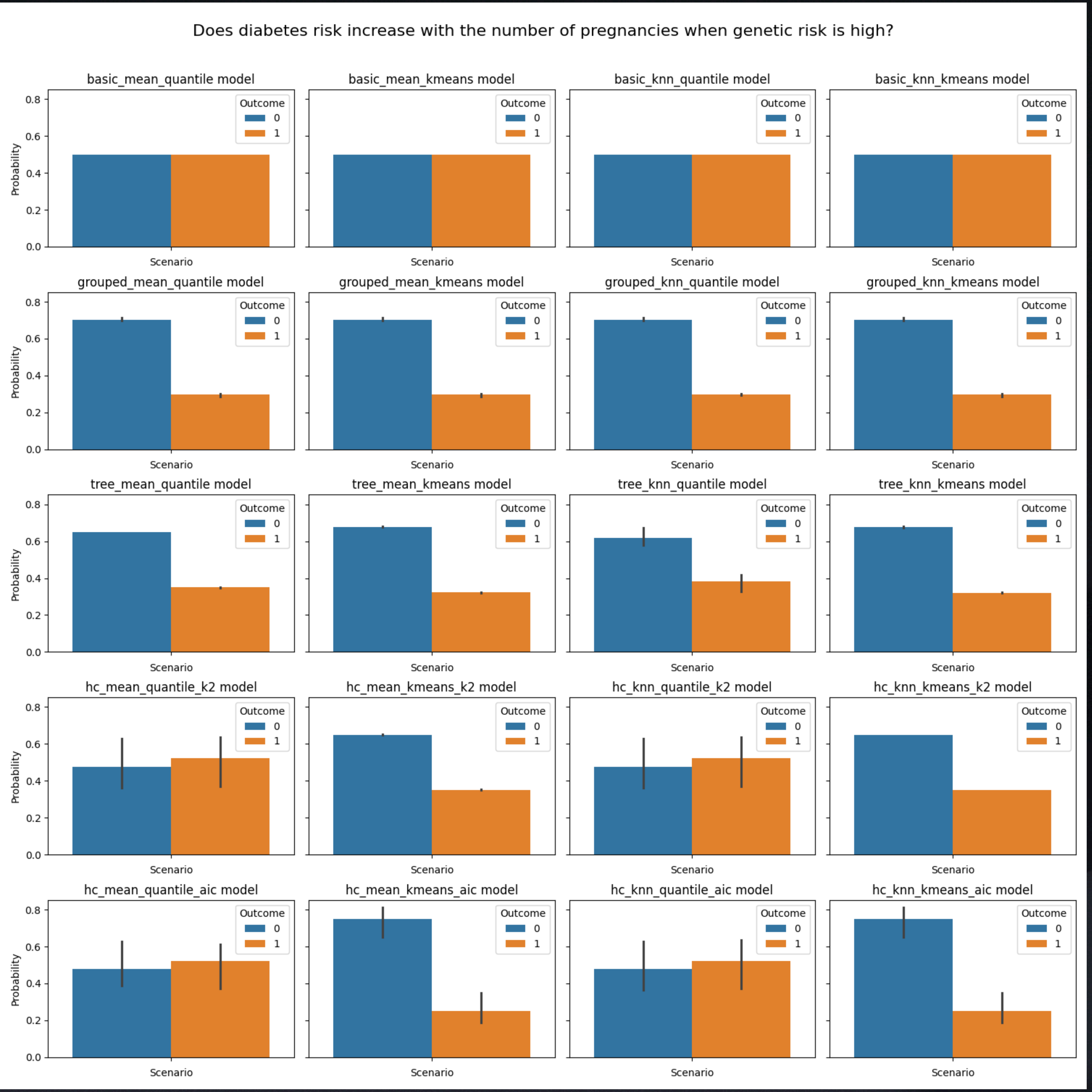


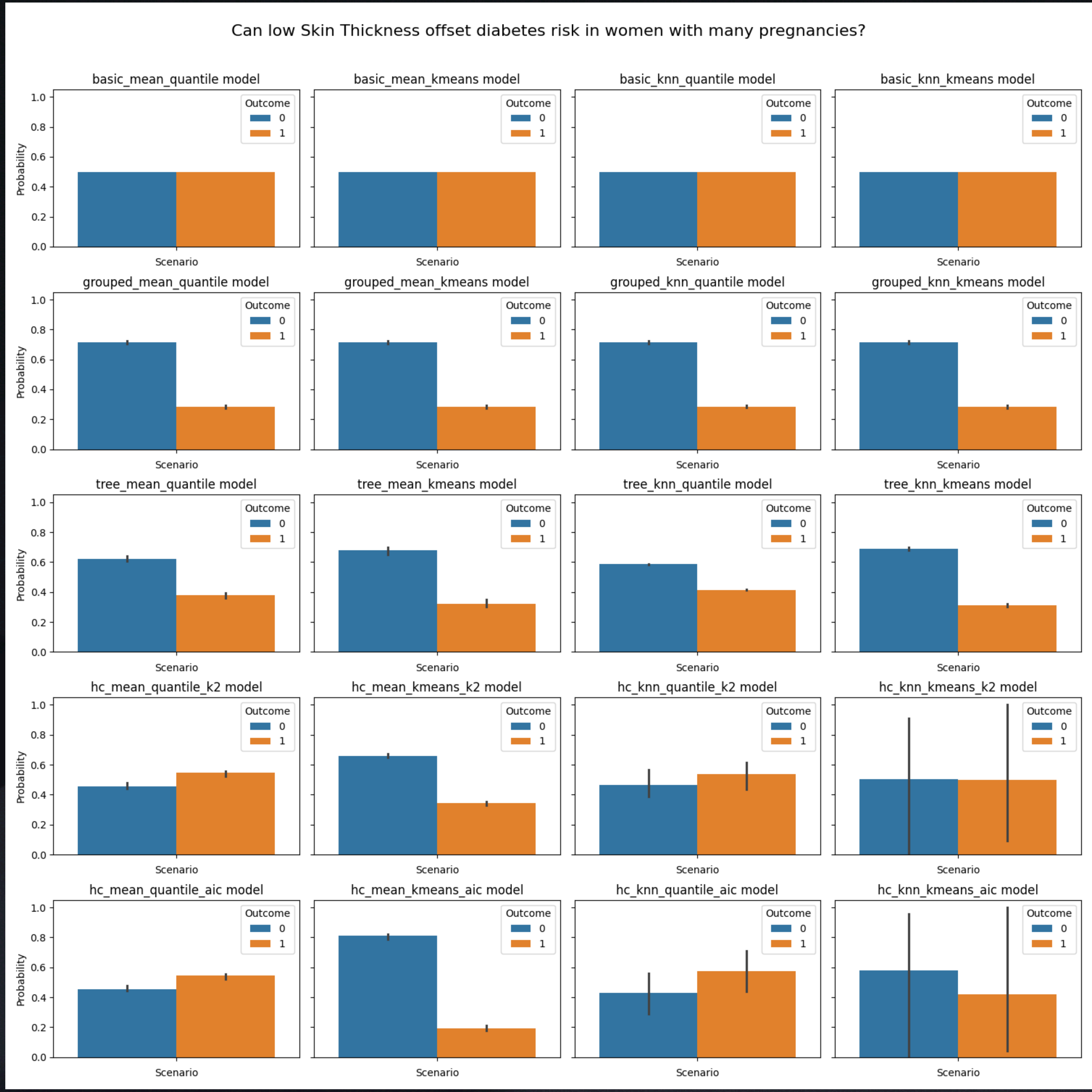
Question 10: Which age group shows the highest sensitivity to BMI changes?



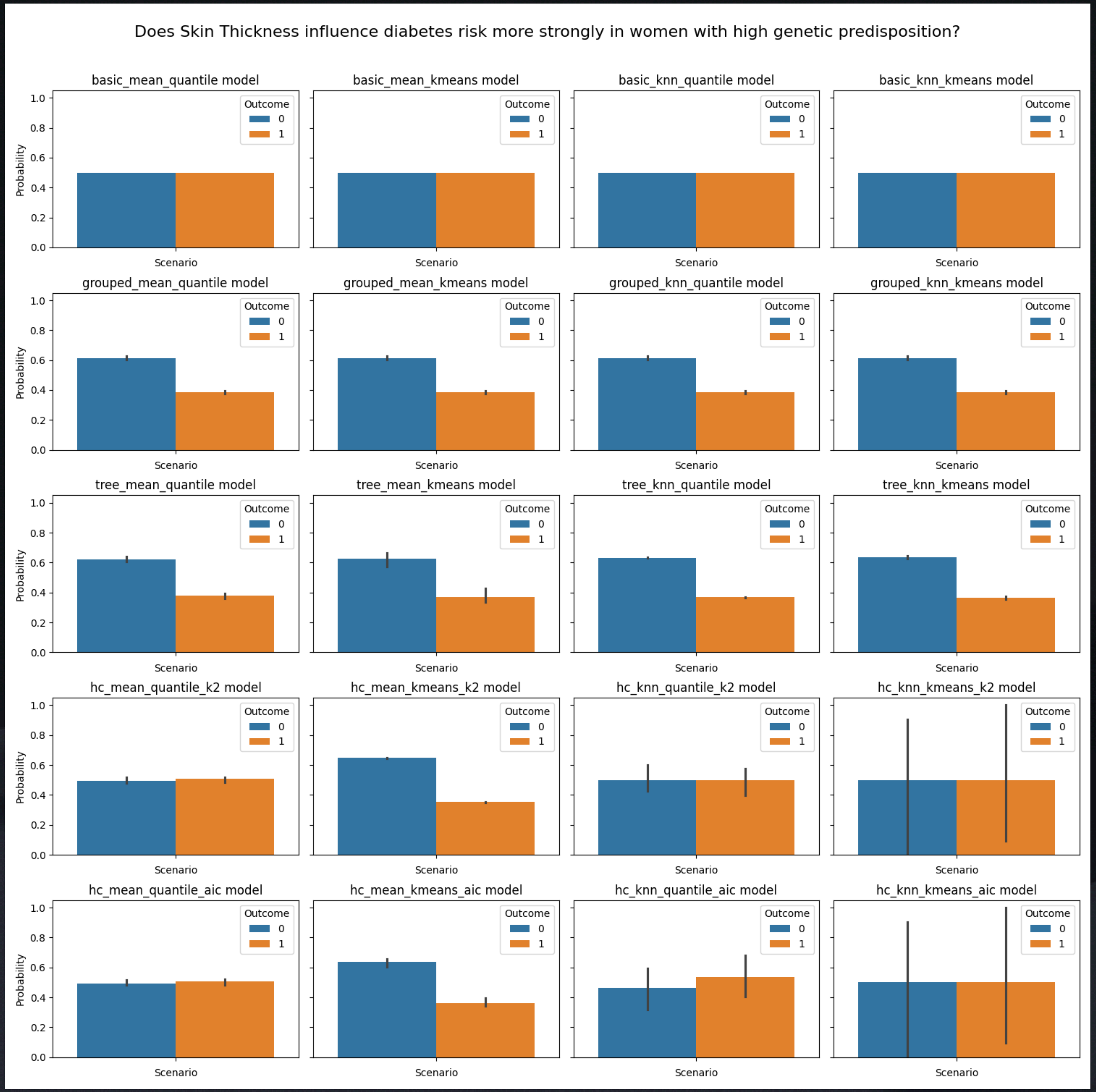
Question 11 : Does diabetes risk increase with the number of pregnancies when genetic risk is high?

Question 12: Can low Skin Thickness offset diabetes risk in women with many pregnancies?

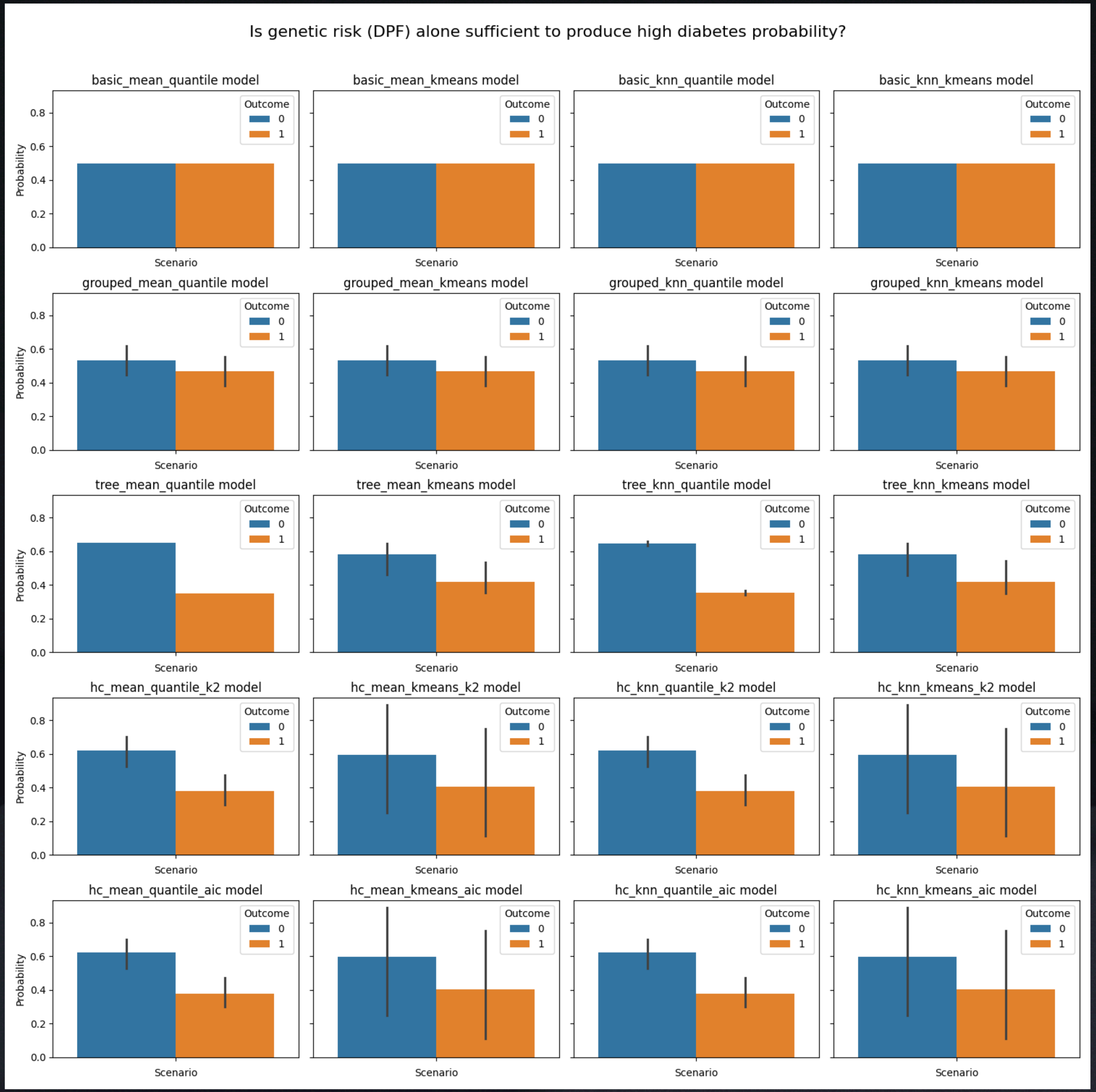




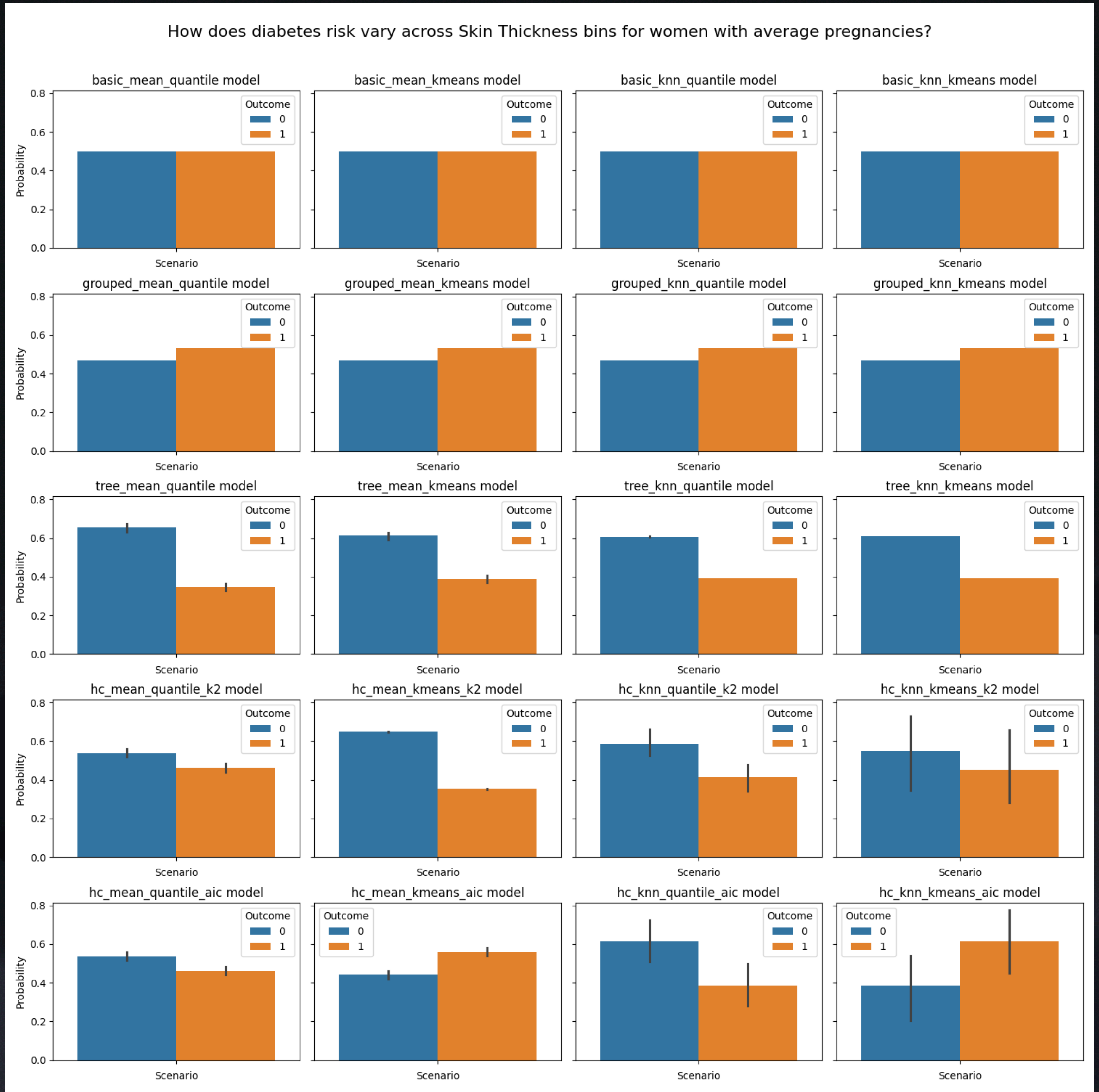
Question 13 : Does Skin Thickness influence diabetes risk more strongly in women with high genetic predisposition?



Question 14: Is genetic risk (DPF) alone sufficient to produce high diabetes probability?



Question15: How does diabetes risk vary across Skin Thickness bins for women with average pregnancies?



Thank You